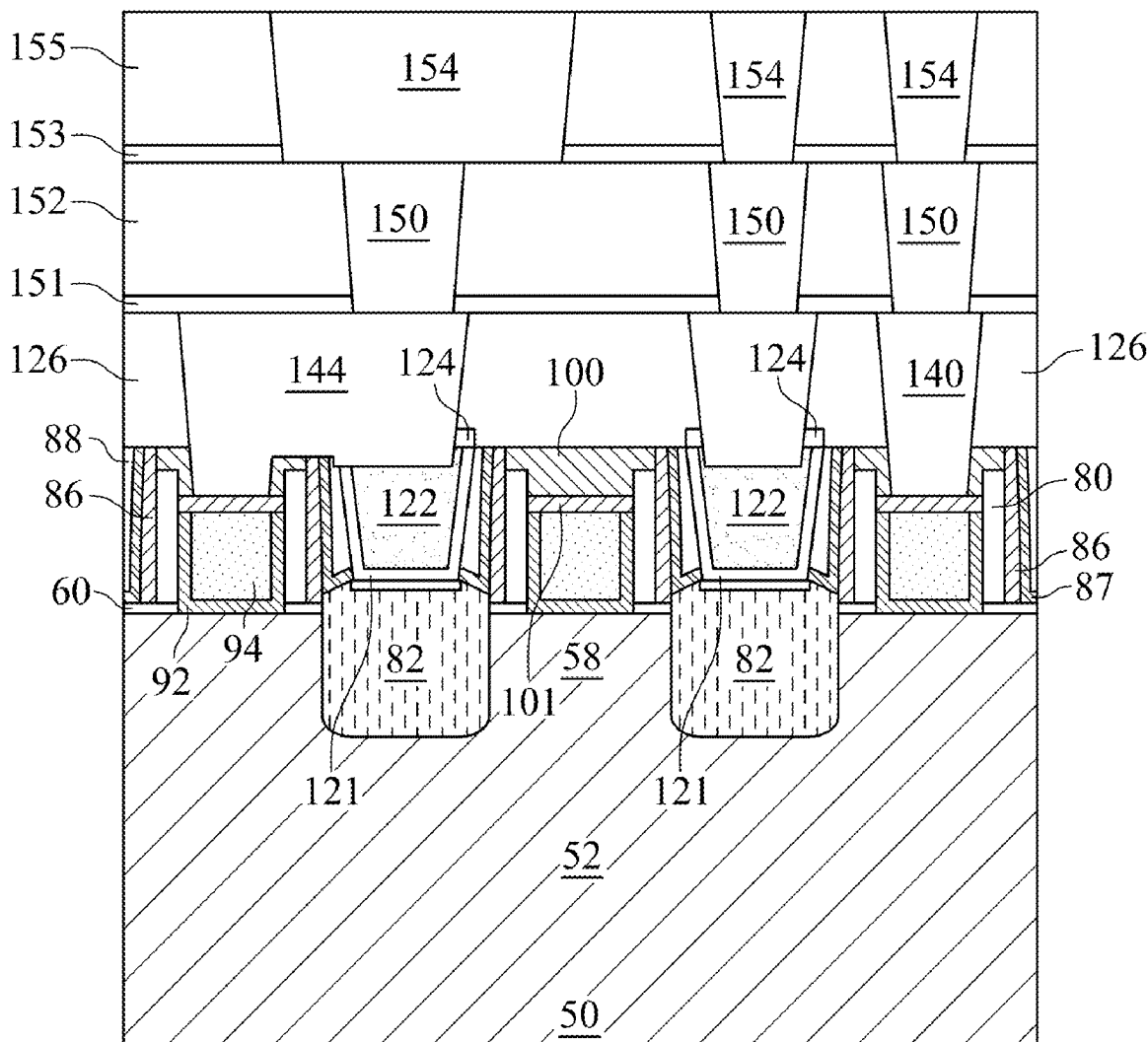


Sep. 4, 2025



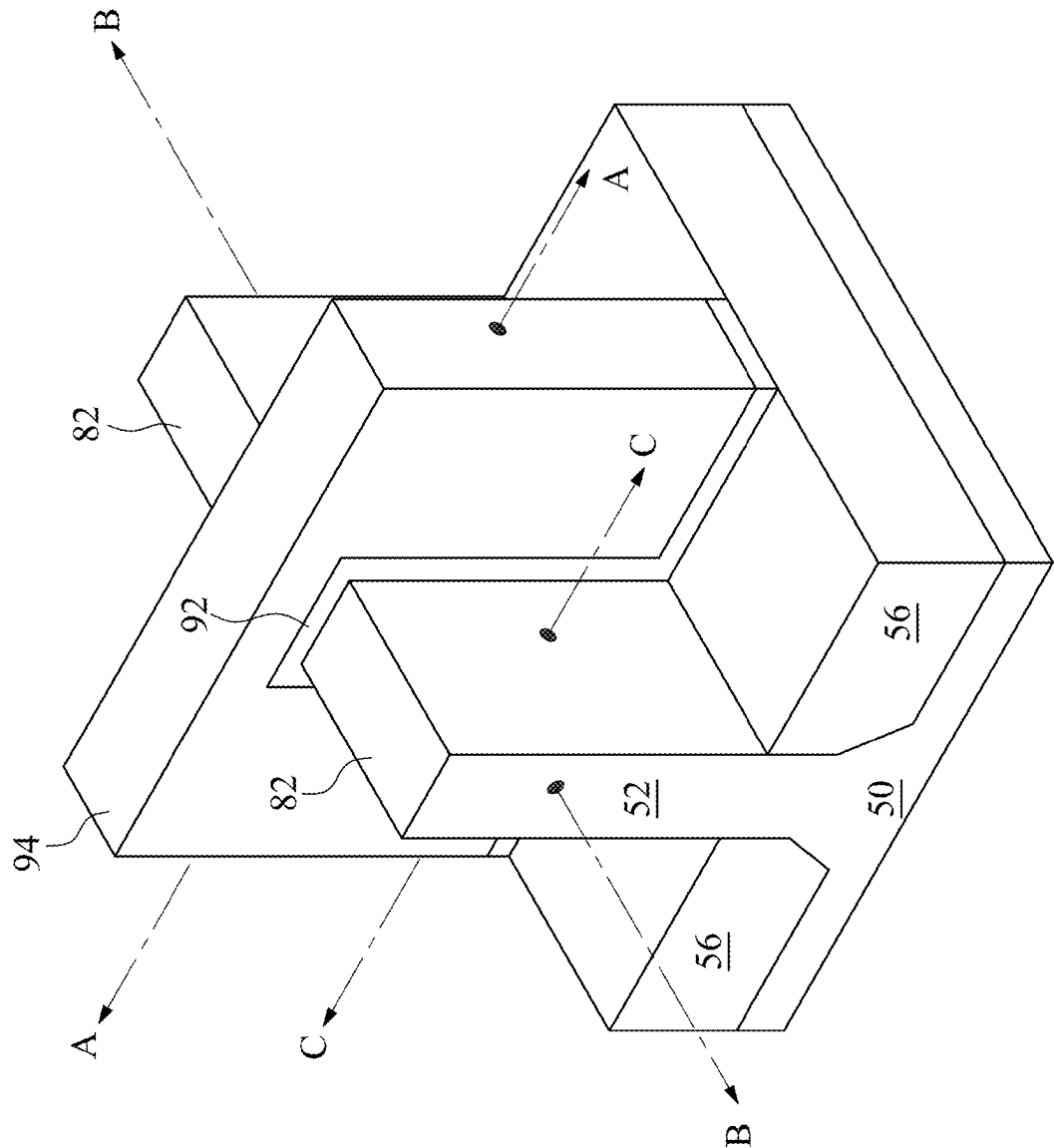


Fig. 1

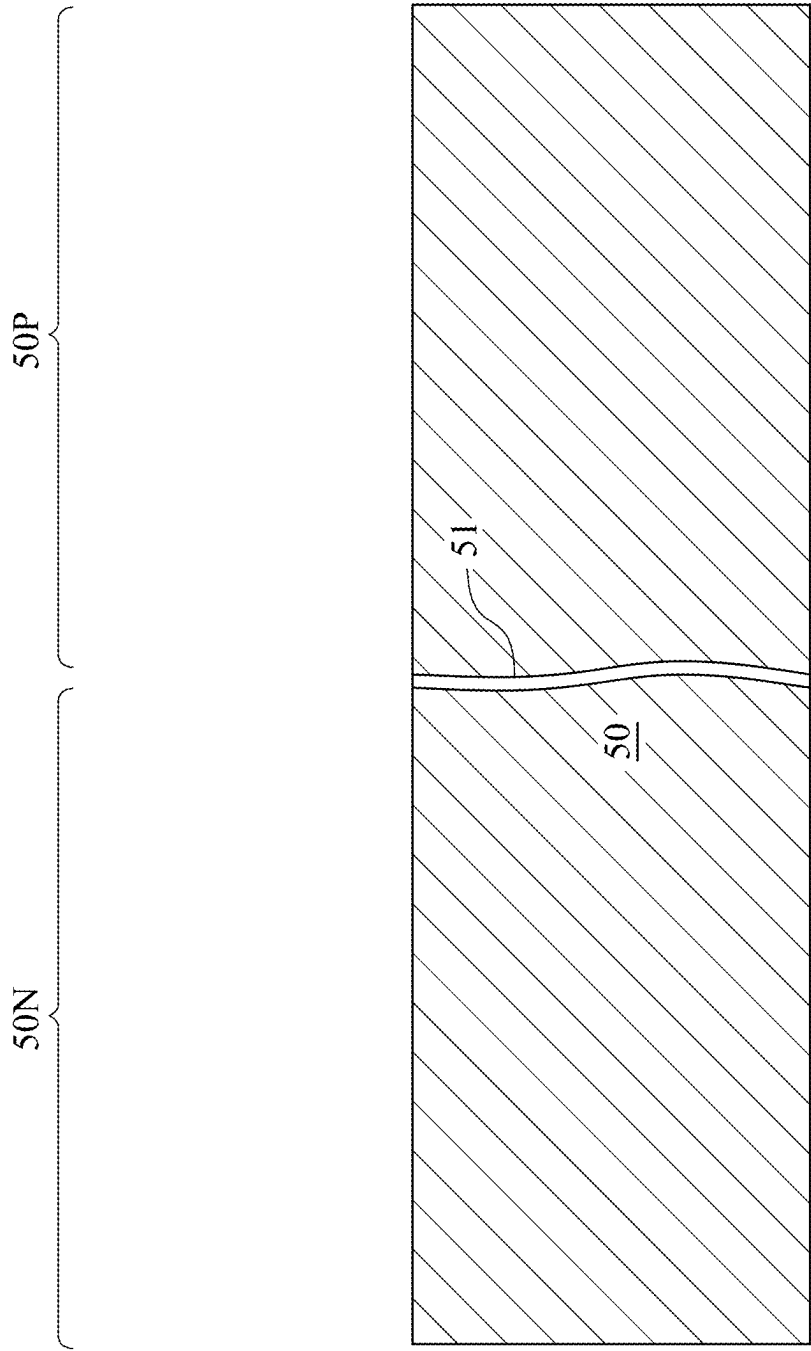


Fig. 2

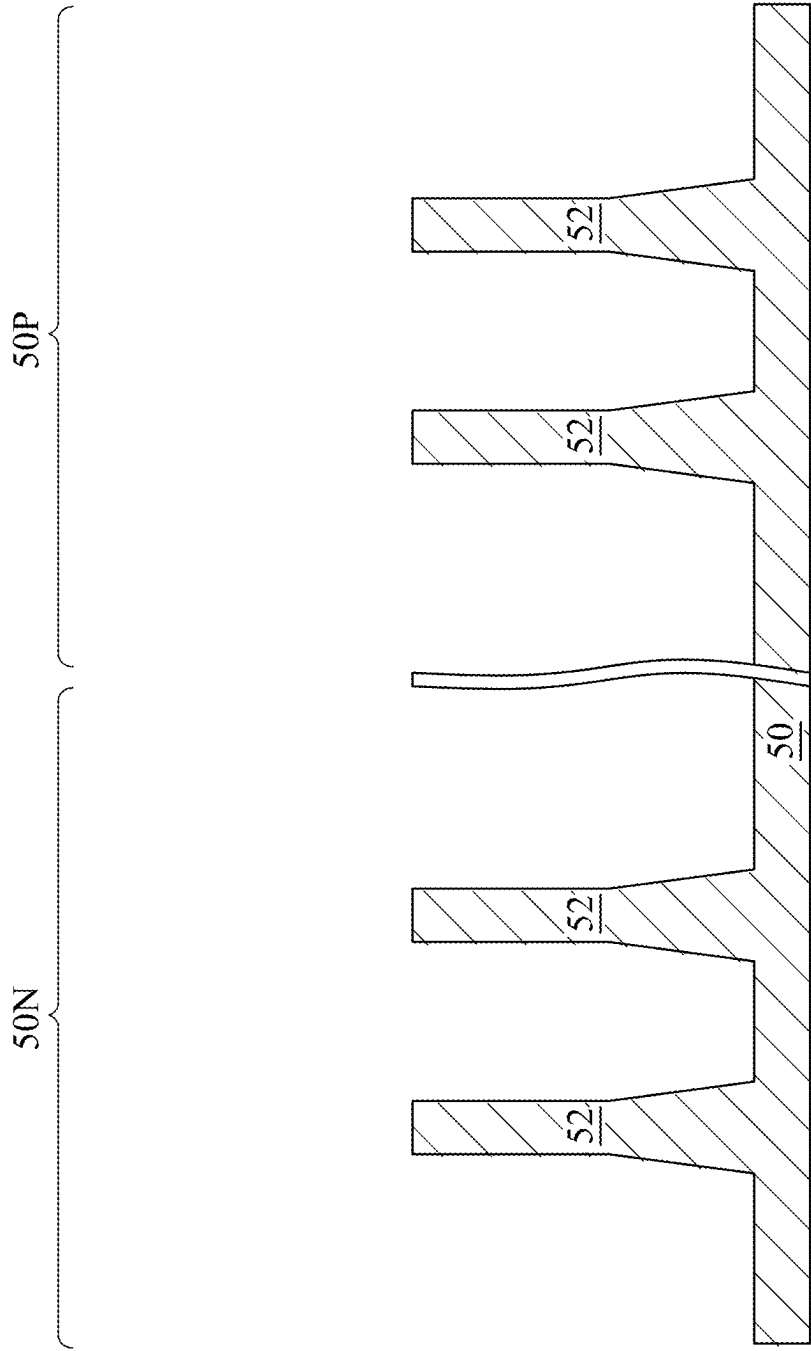


Fig. 3

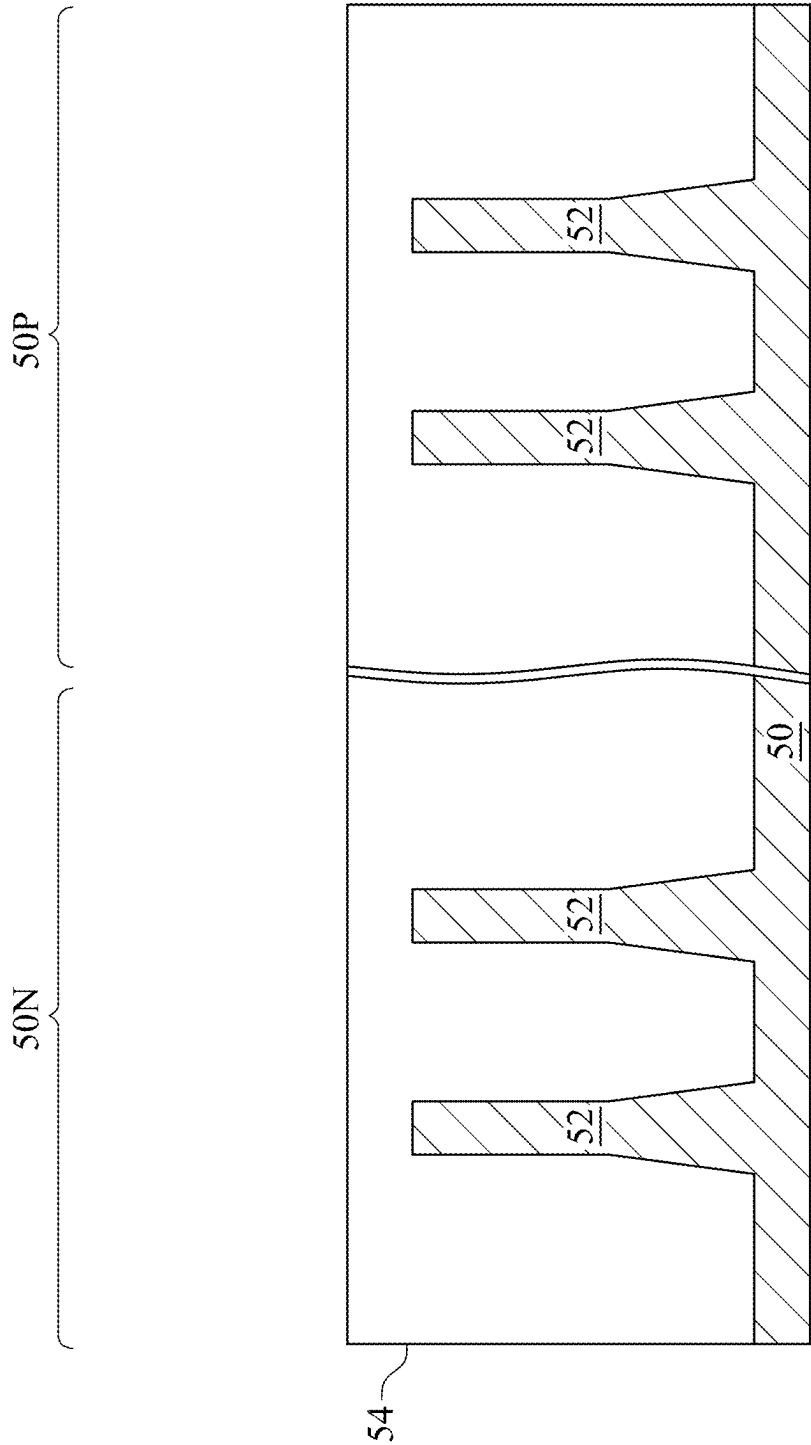


Fig. 4

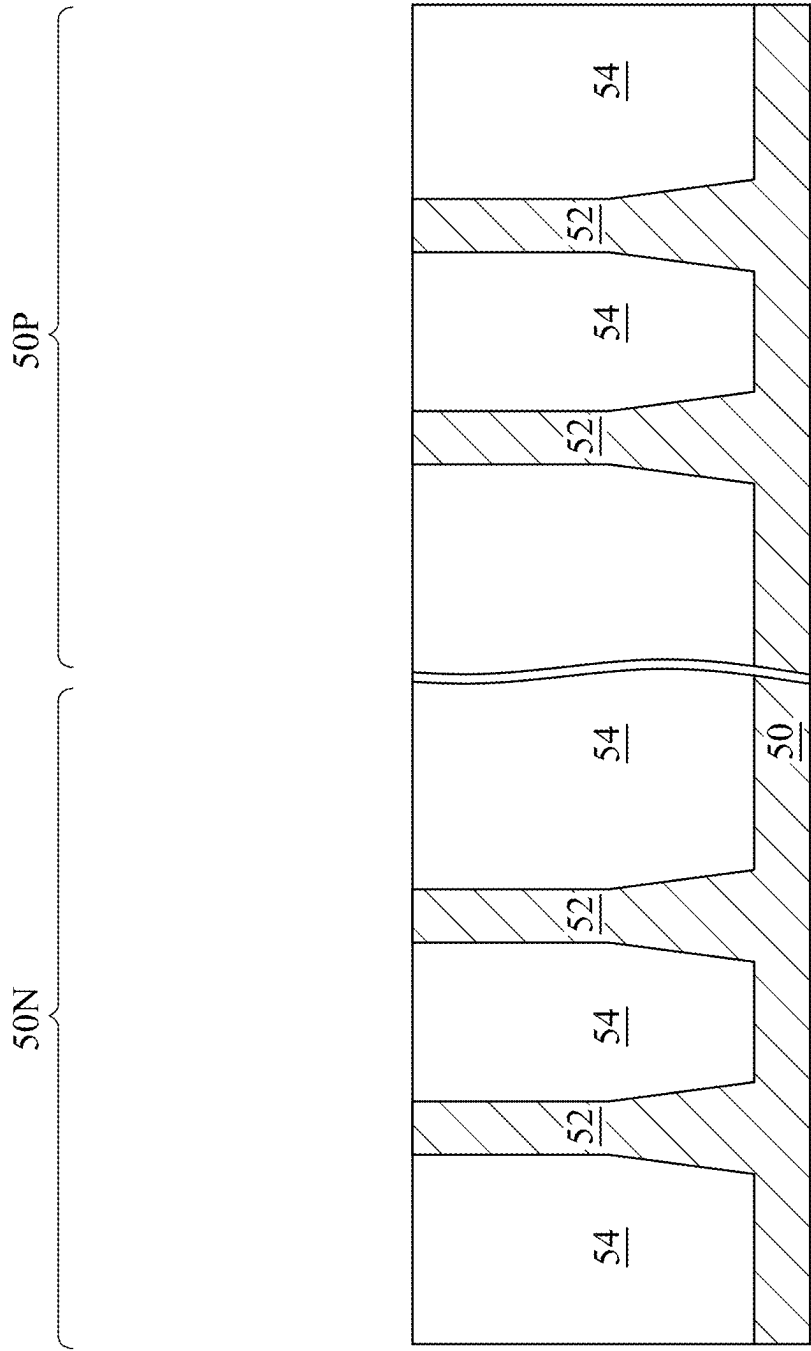


Fig. 5

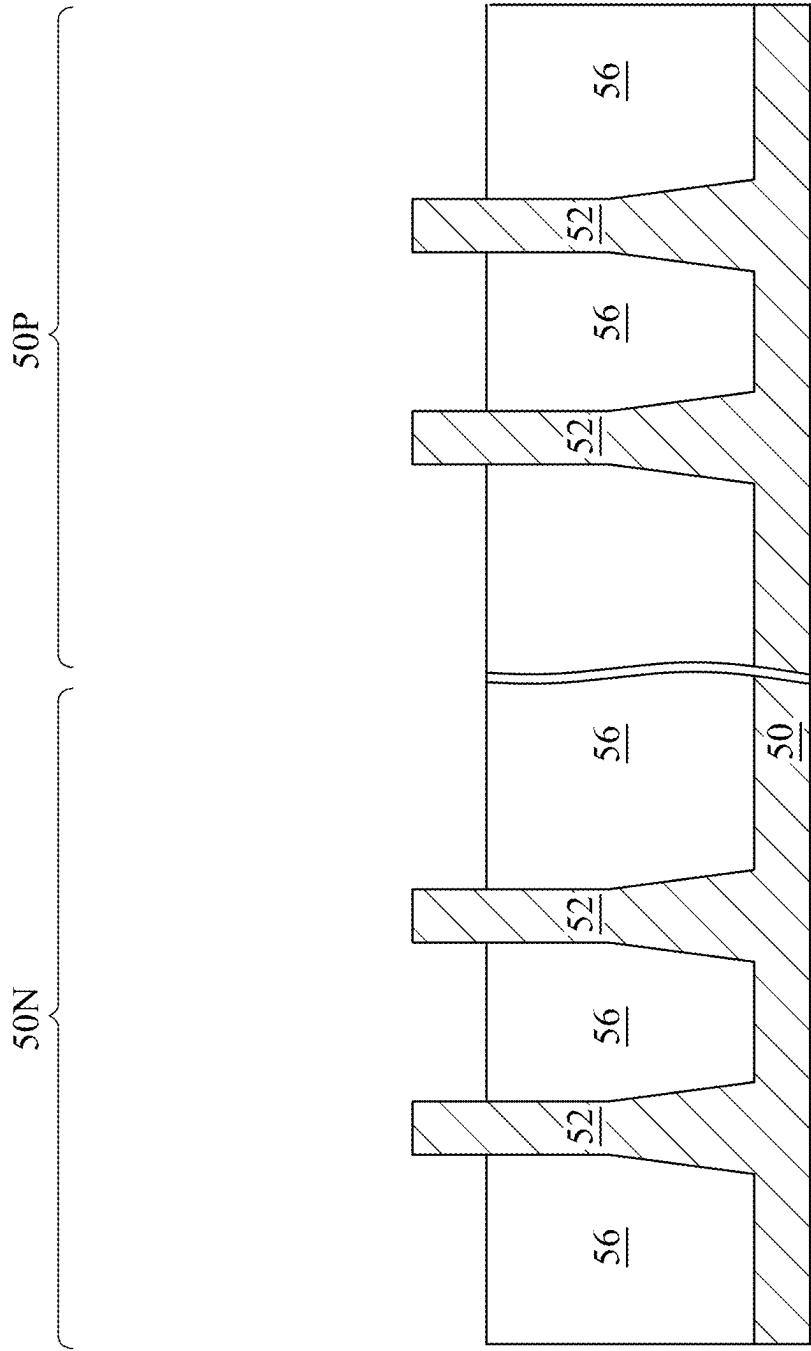


Fig. 6

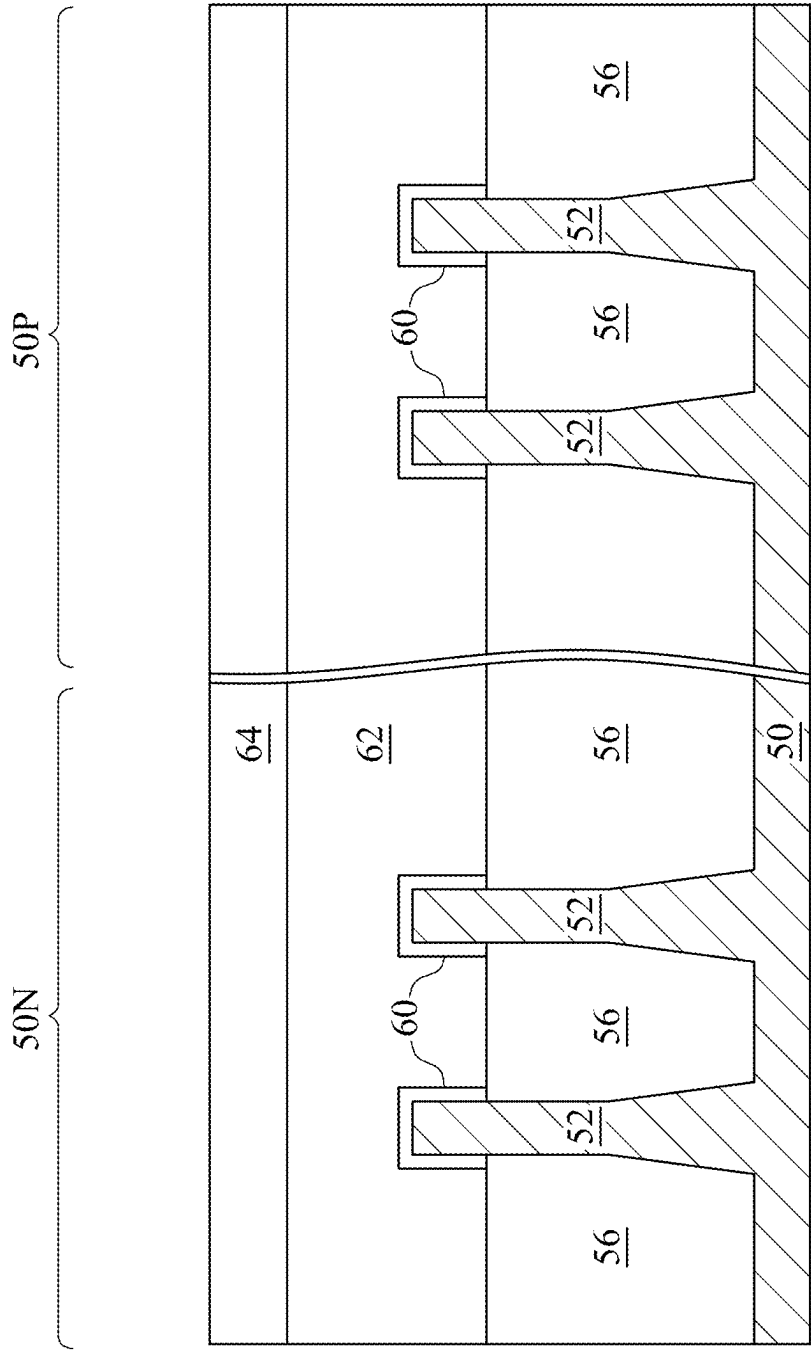


Fig. 7

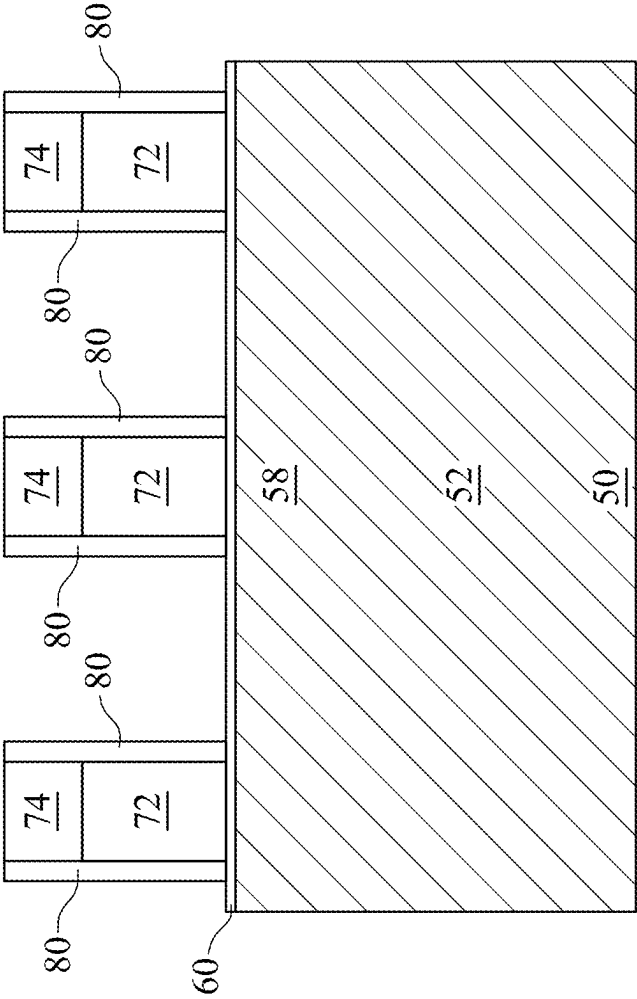


Fig. 8A

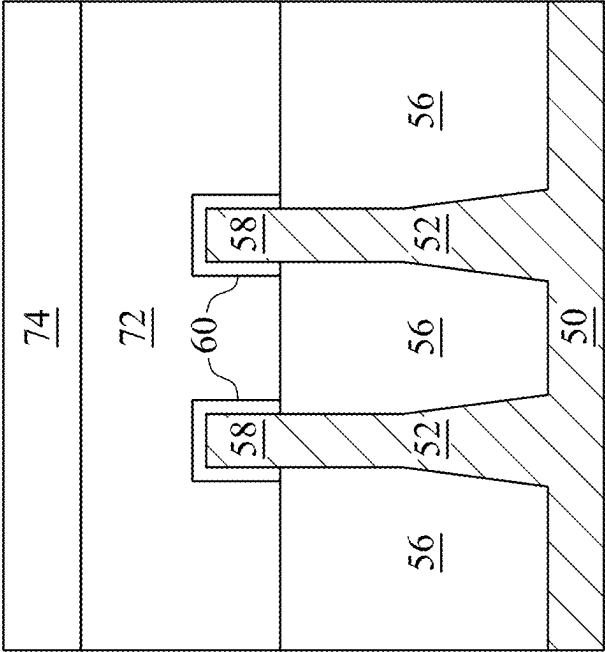


Fig. 8B

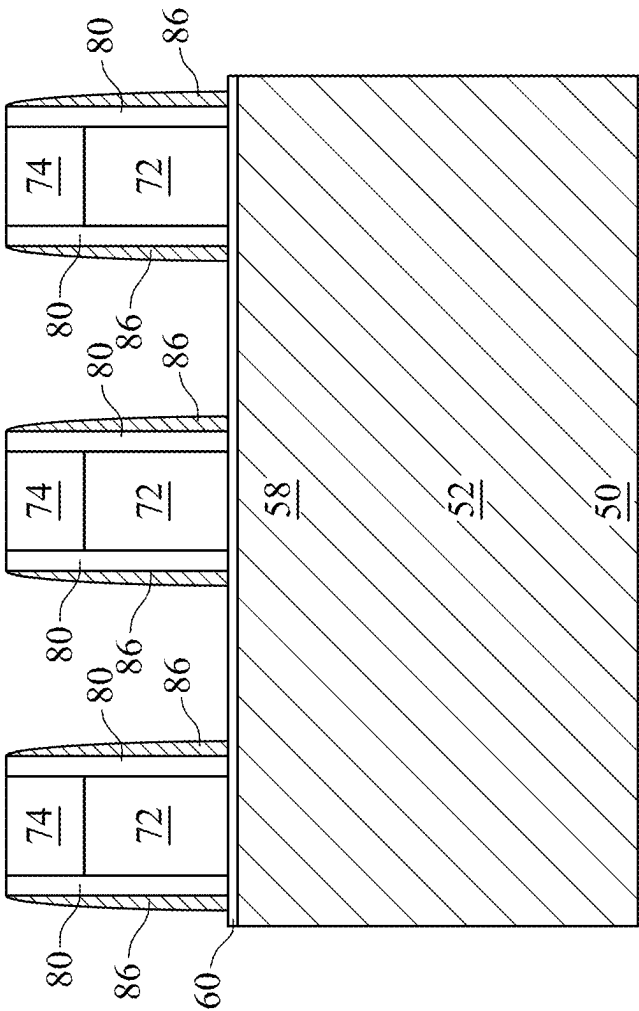


Fig. 9A

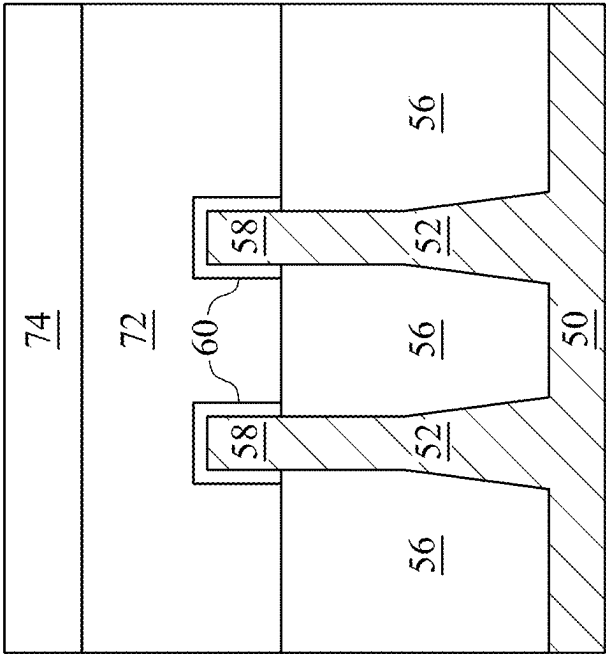


Fig. 9B

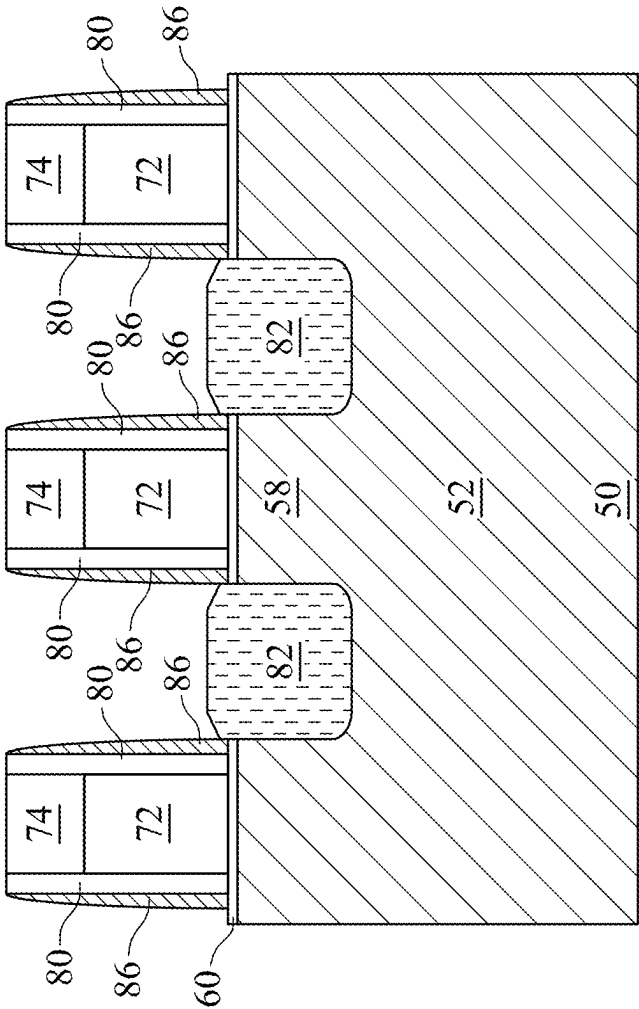


Fig. 10A

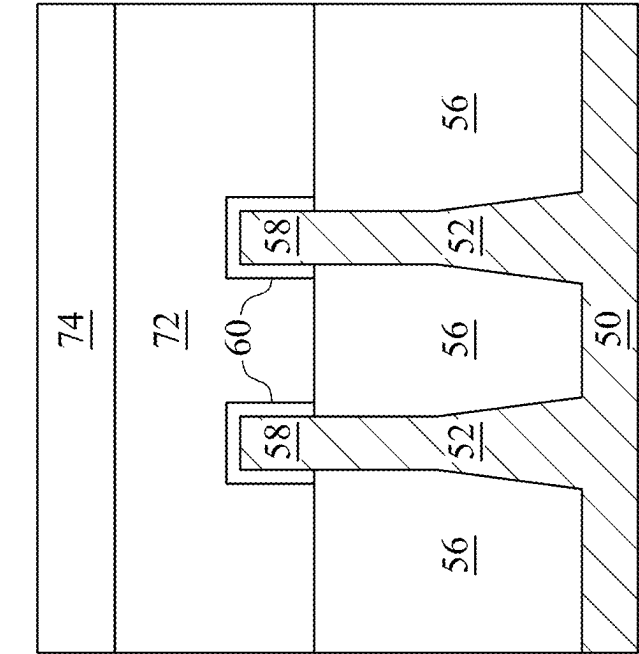


Fig. 10B

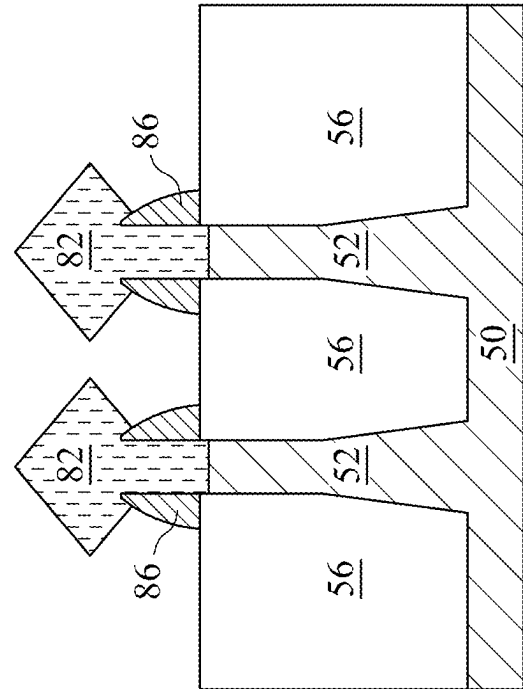


Fig. 10D

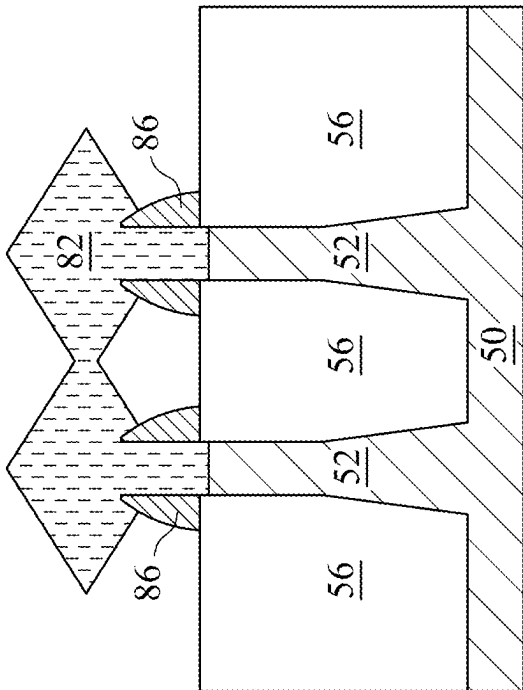


Fig. 10C

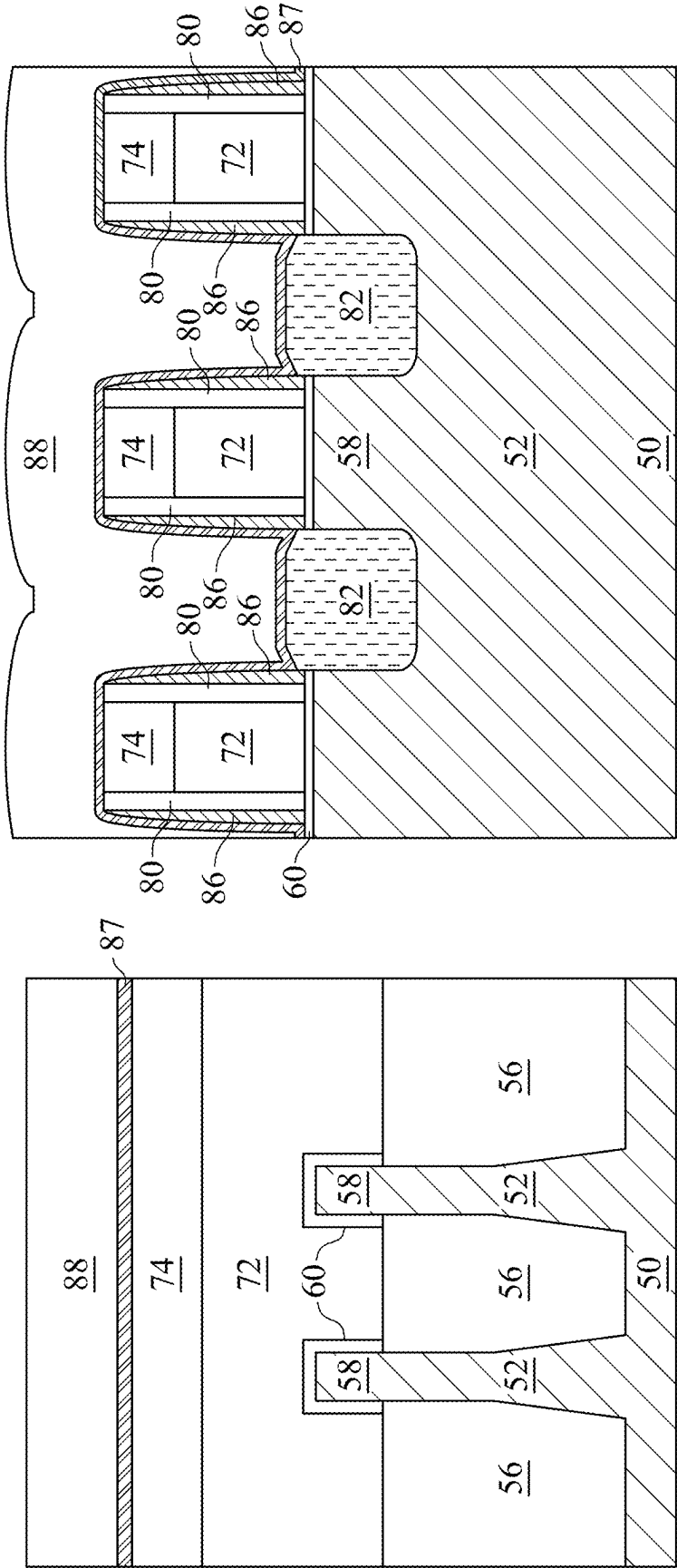


Fig. 11A

Fig. 11B

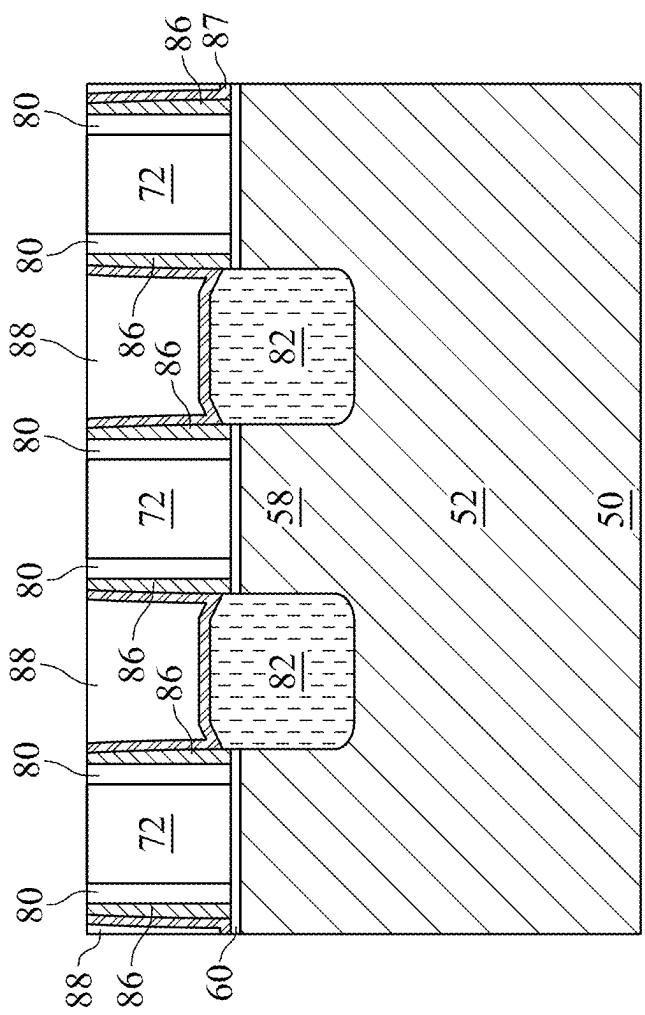


Fig. 12A

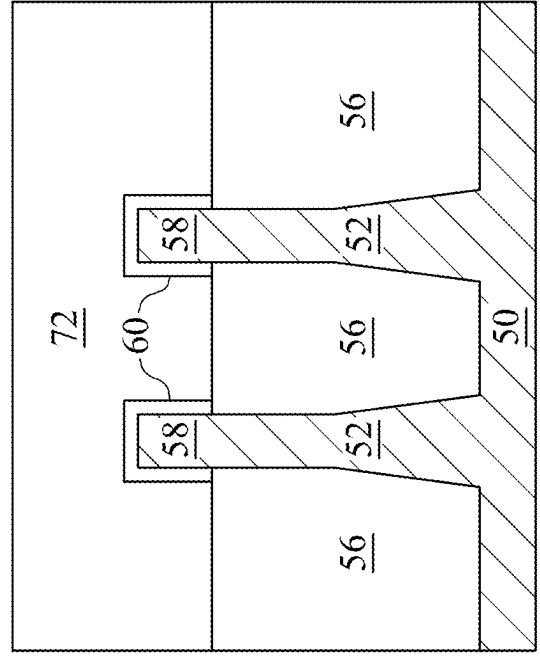


Fig. 12B

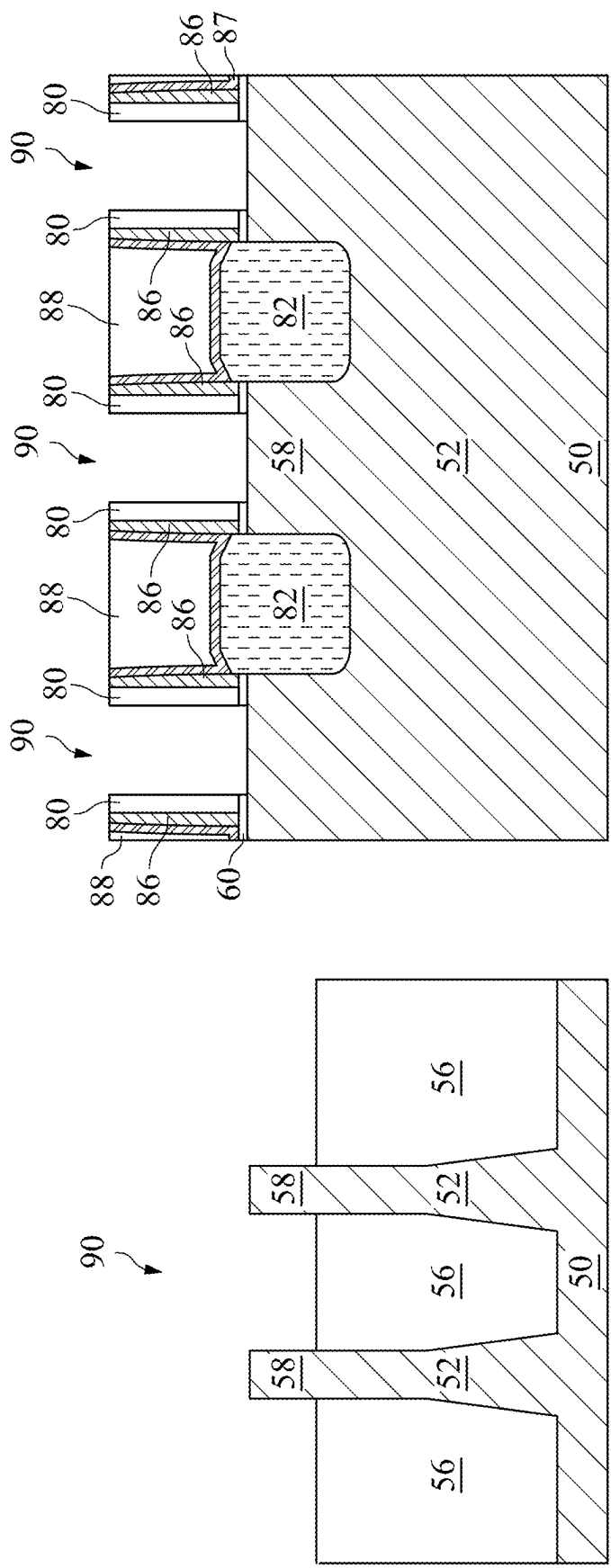


Fig. 13A

Fig. 13B

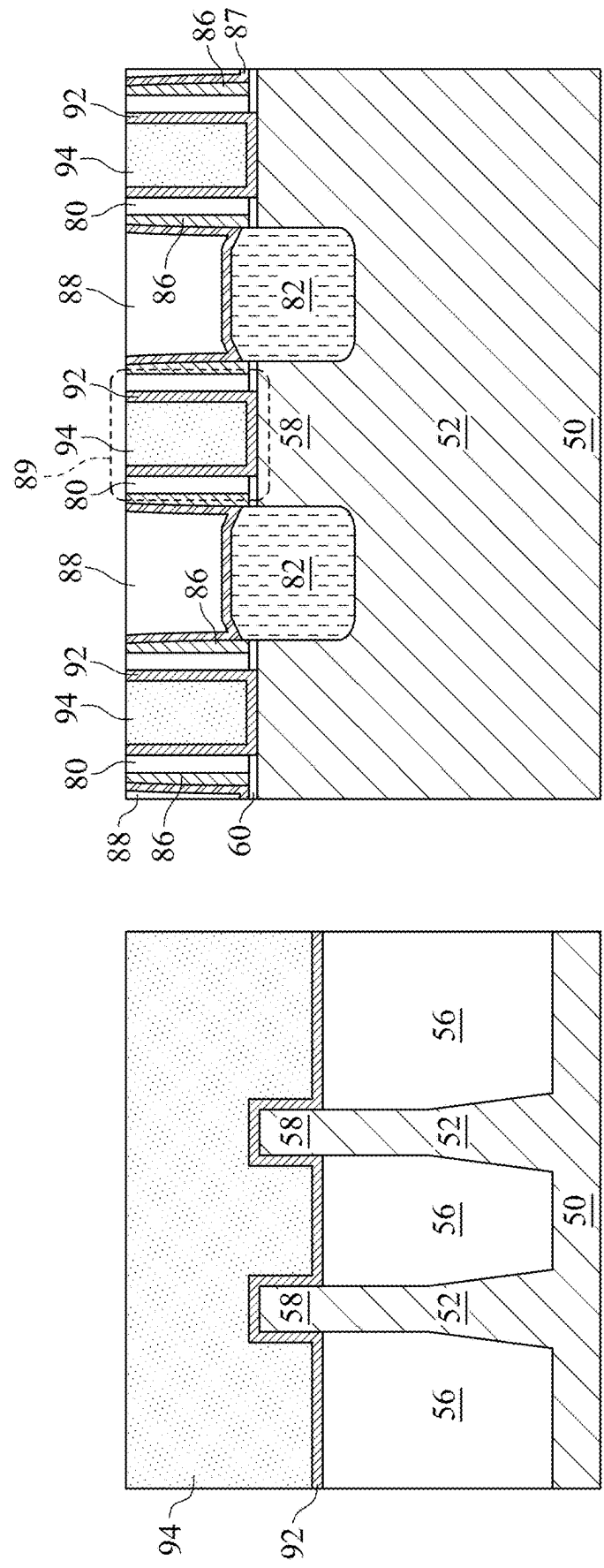


Fig. 14A

Fig. 14B

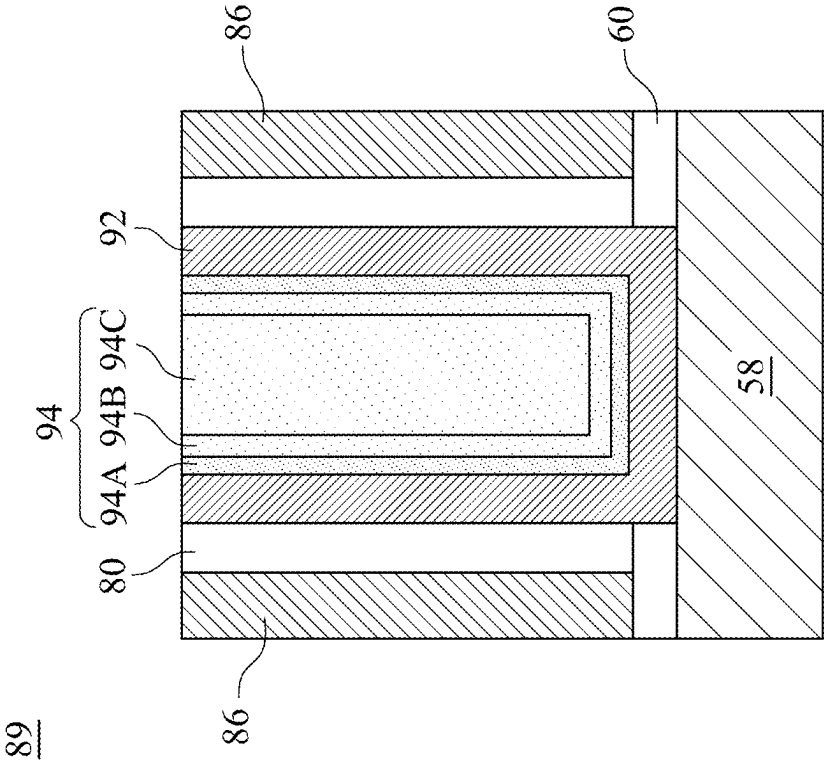


Fig. 14C

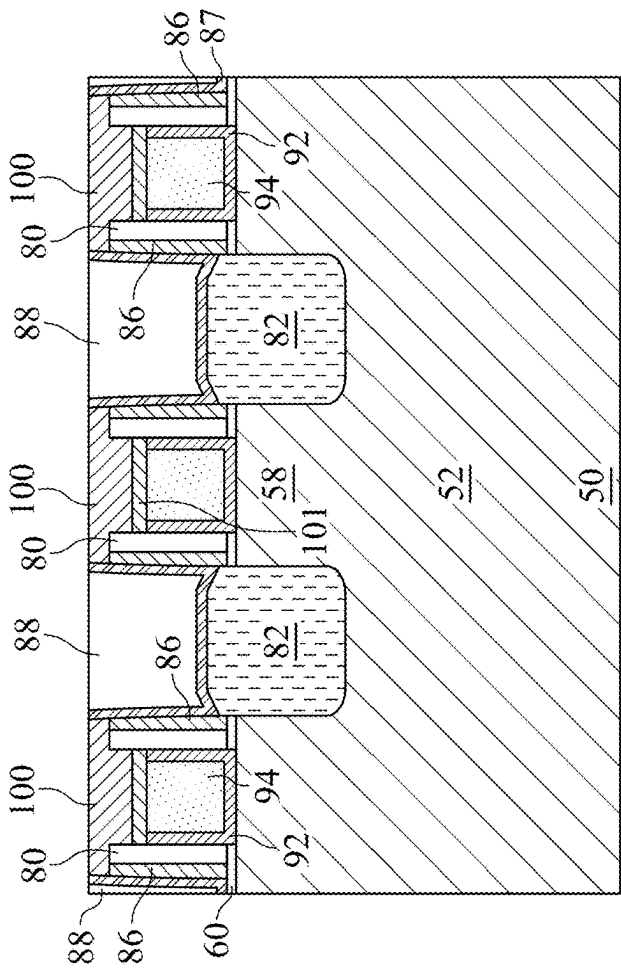


Fig. 15B

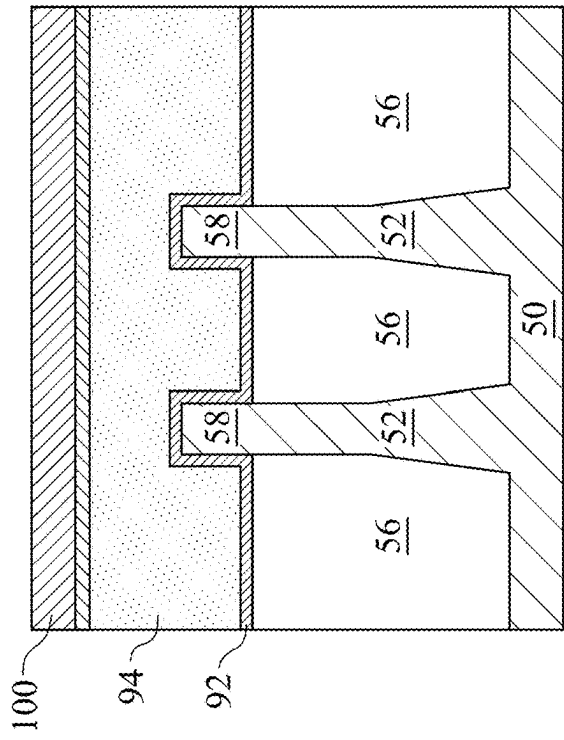


Fig. 15A

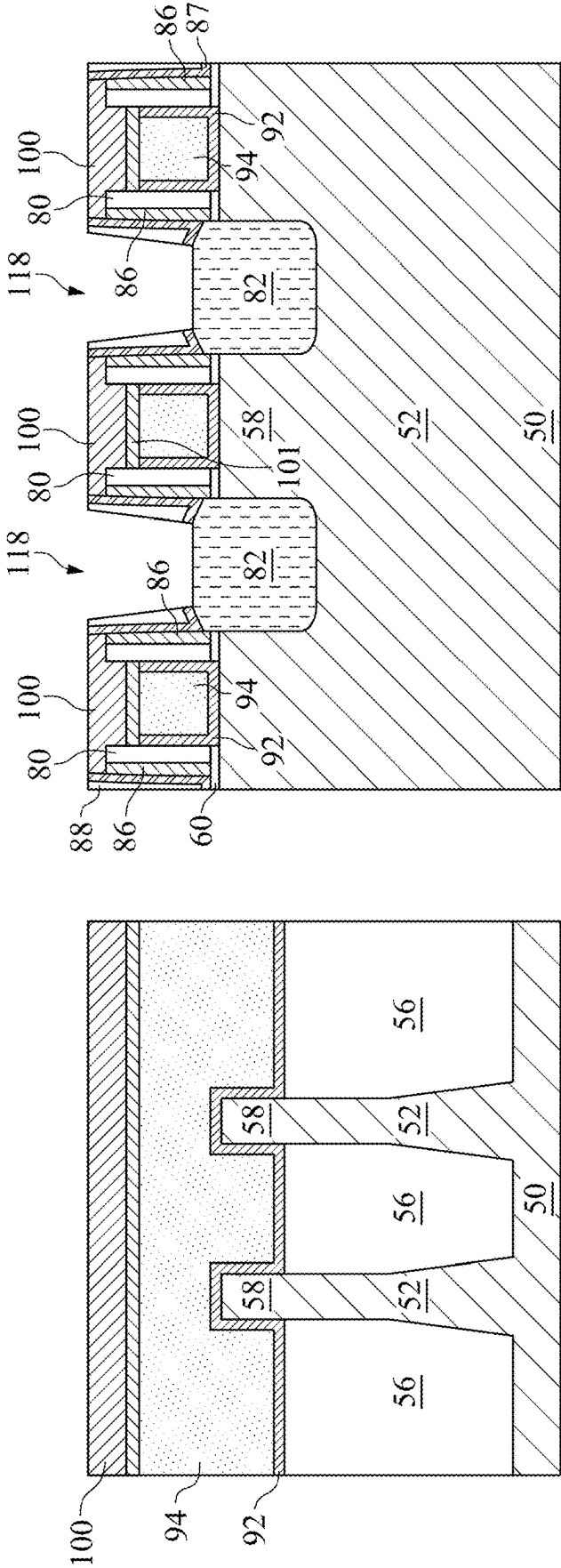


Fig. 16A

Fig. 16B

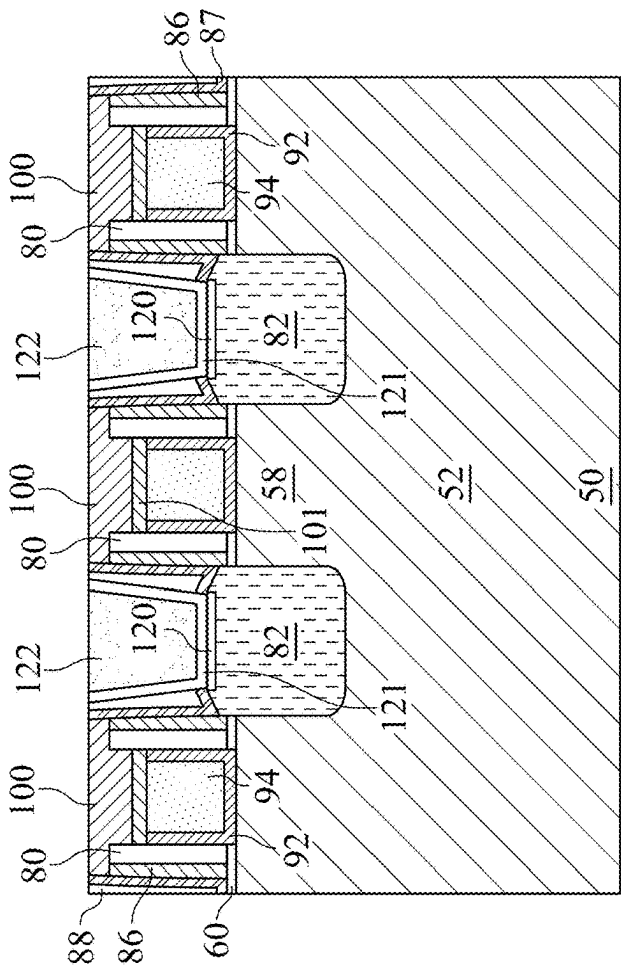


Fig. 17B

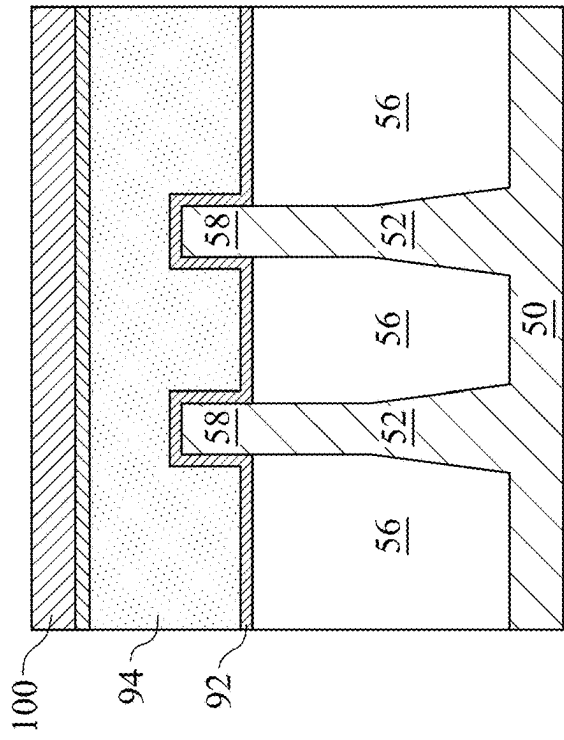


Fig. 17A

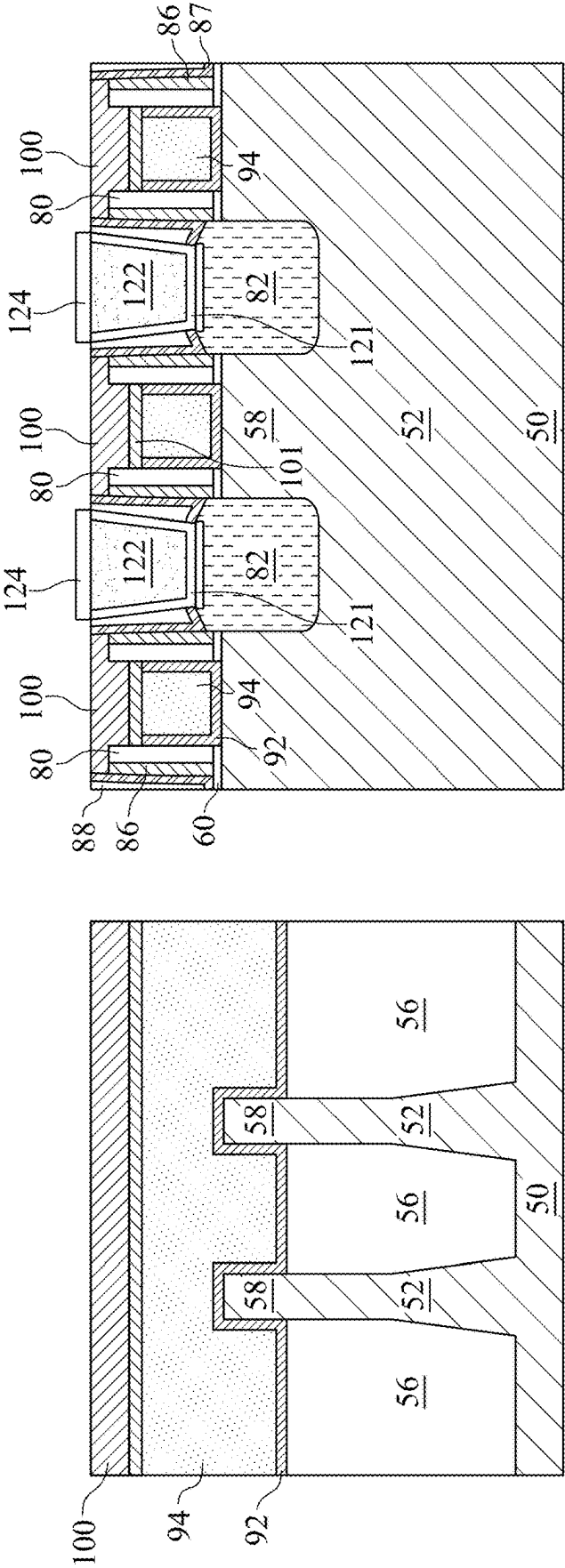


Fig. 18A

Fig. 18B

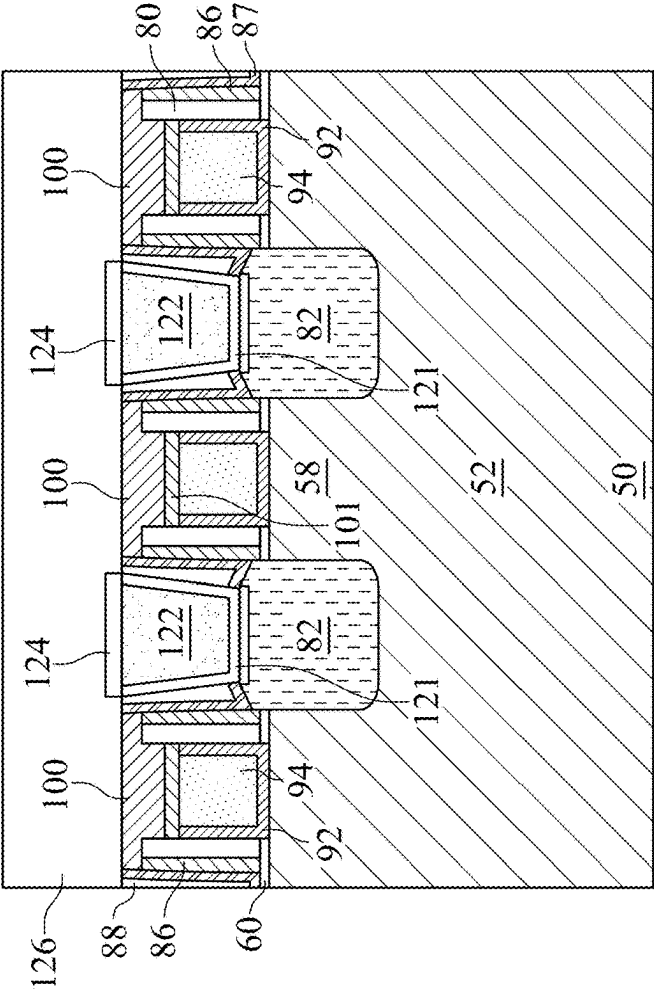


Fig. 19B

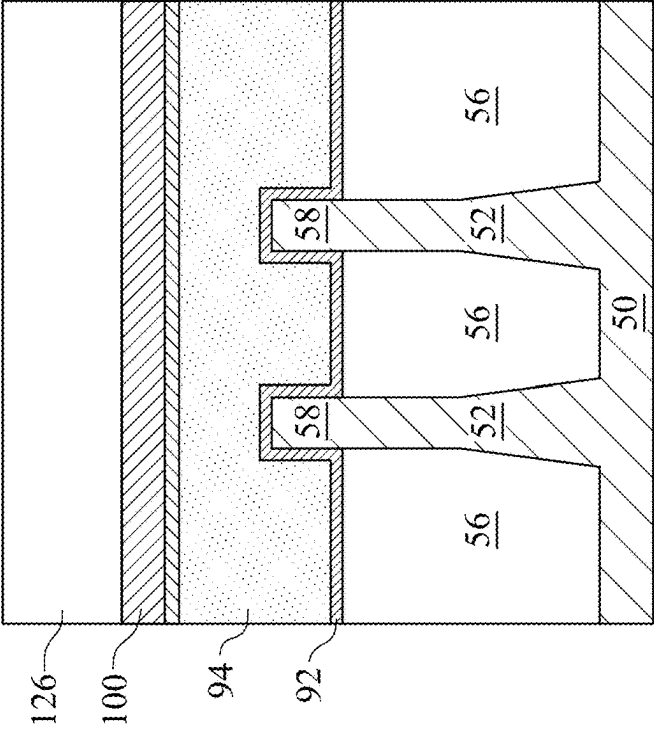


Fig. 19A

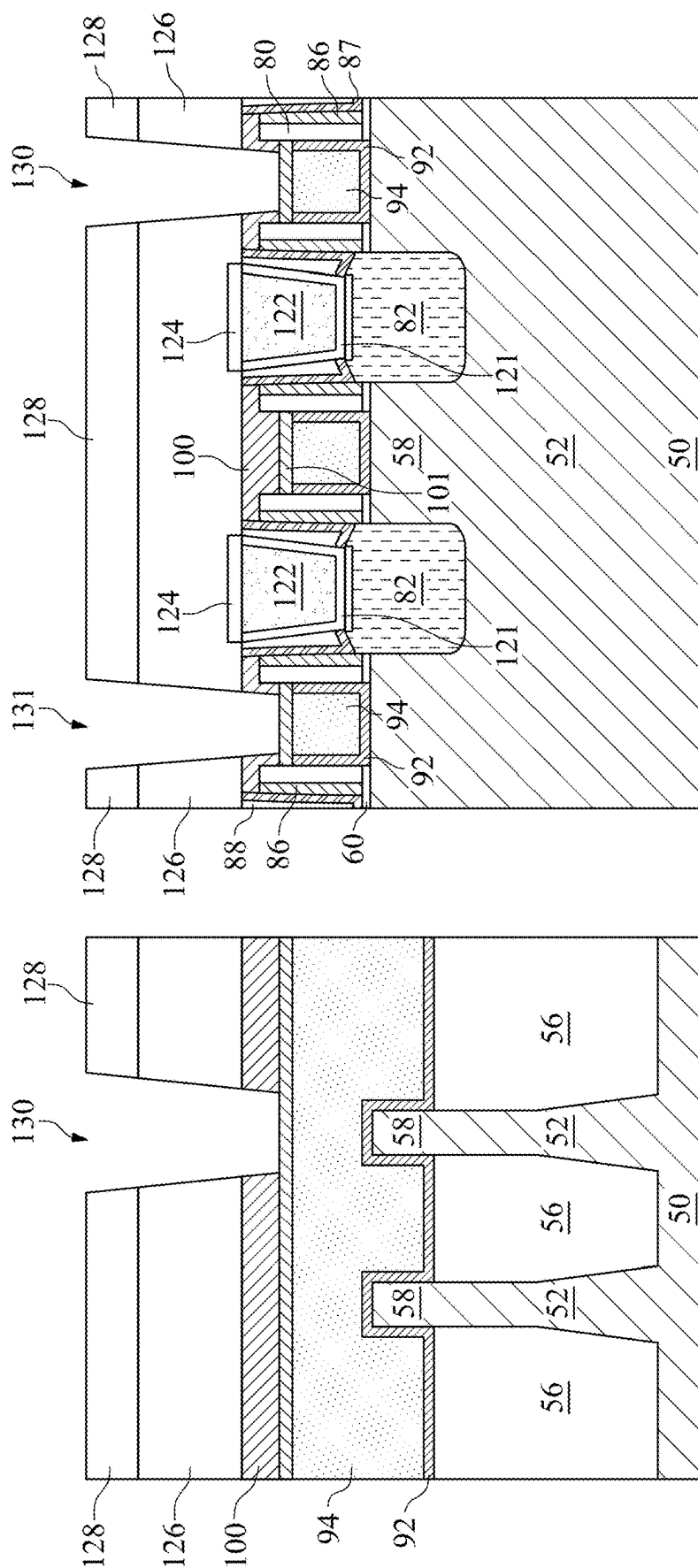


Fig. 20A

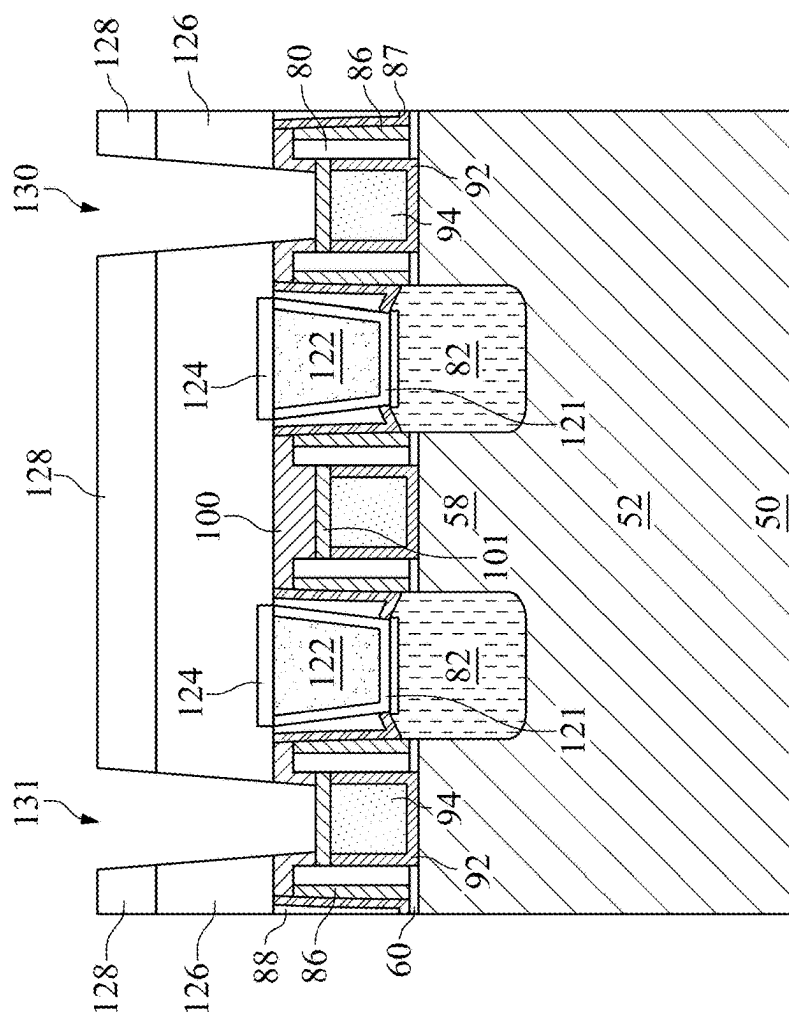


Fig. 20B

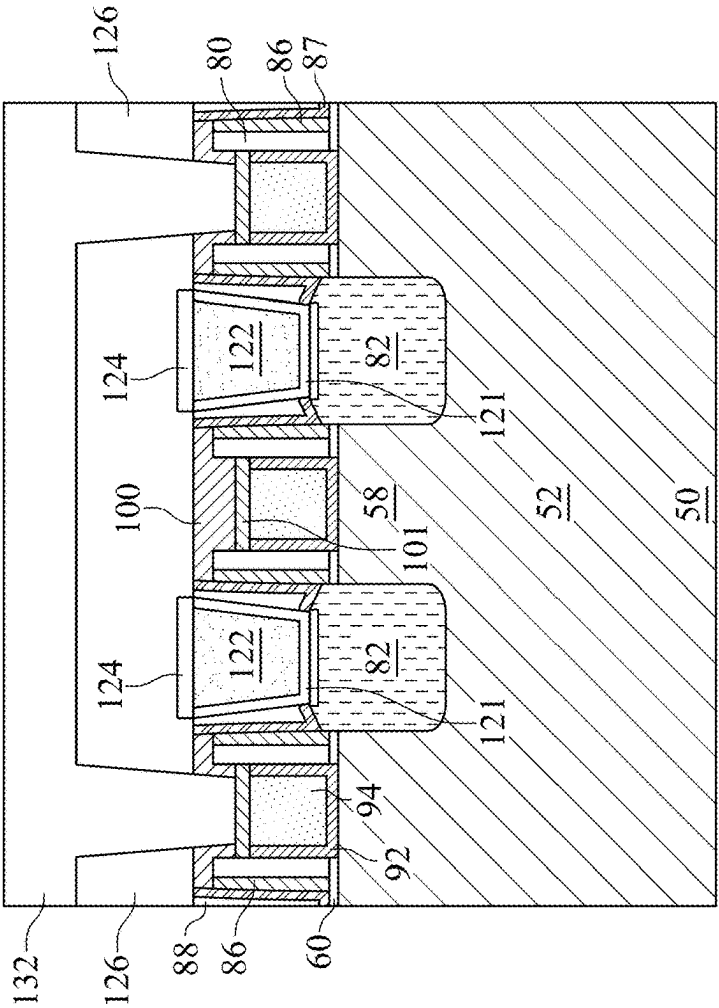


Fig. 21B

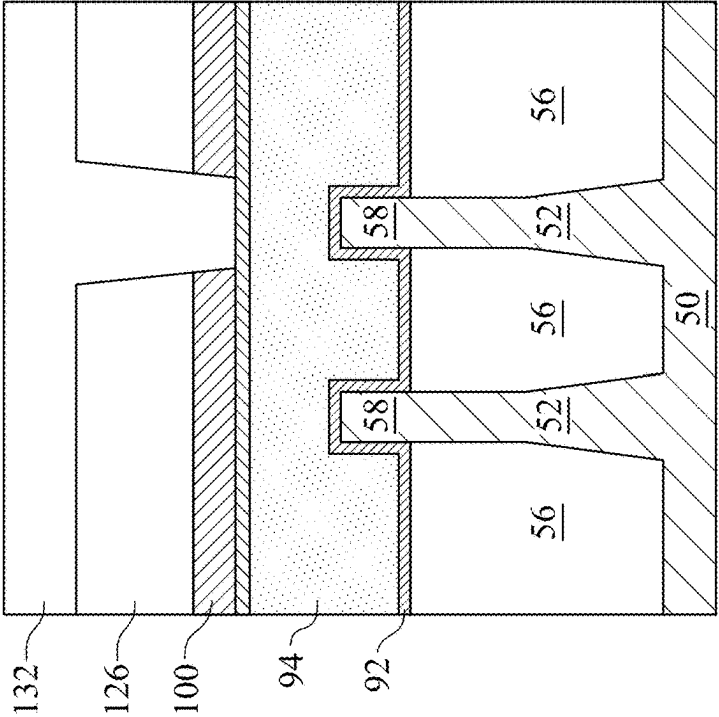


Fig. 21A

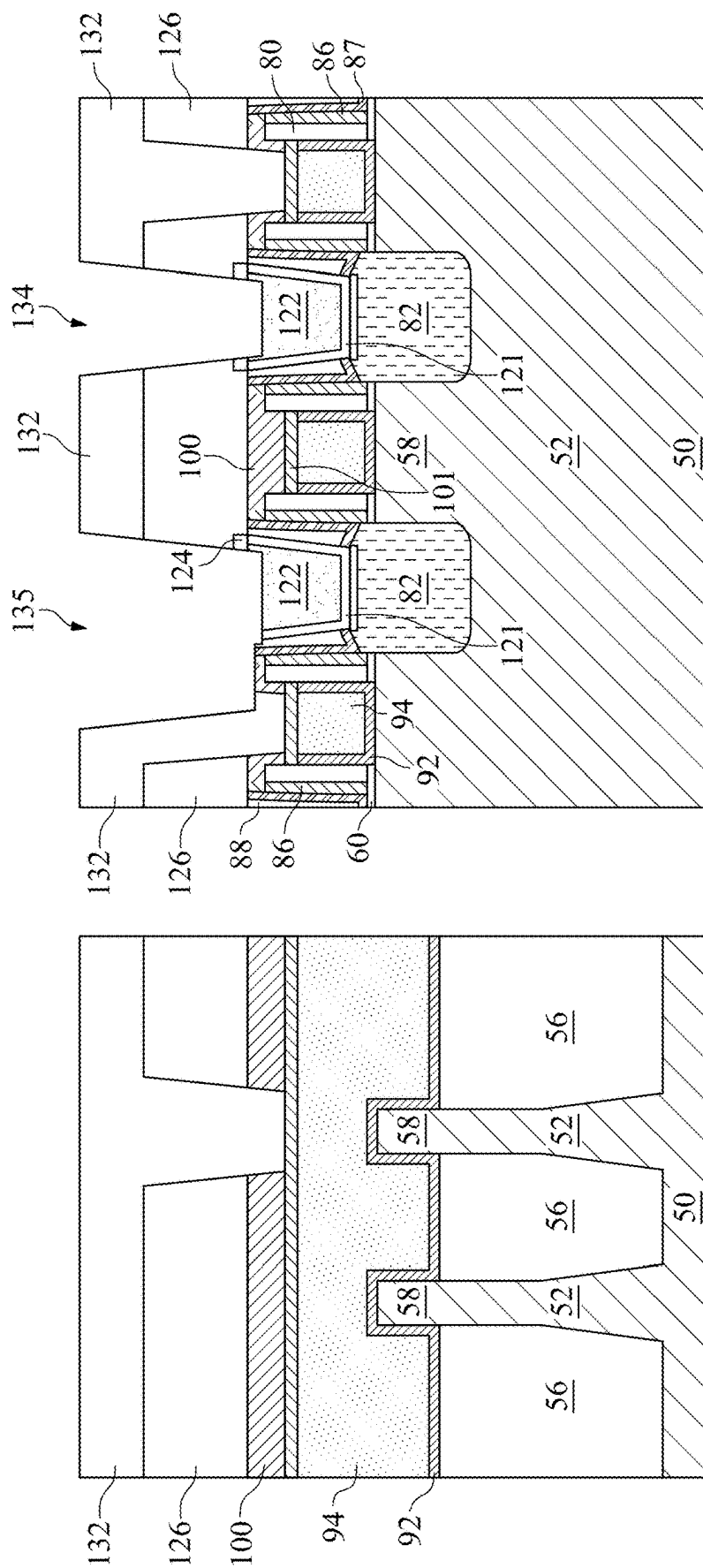


Fig. 22A

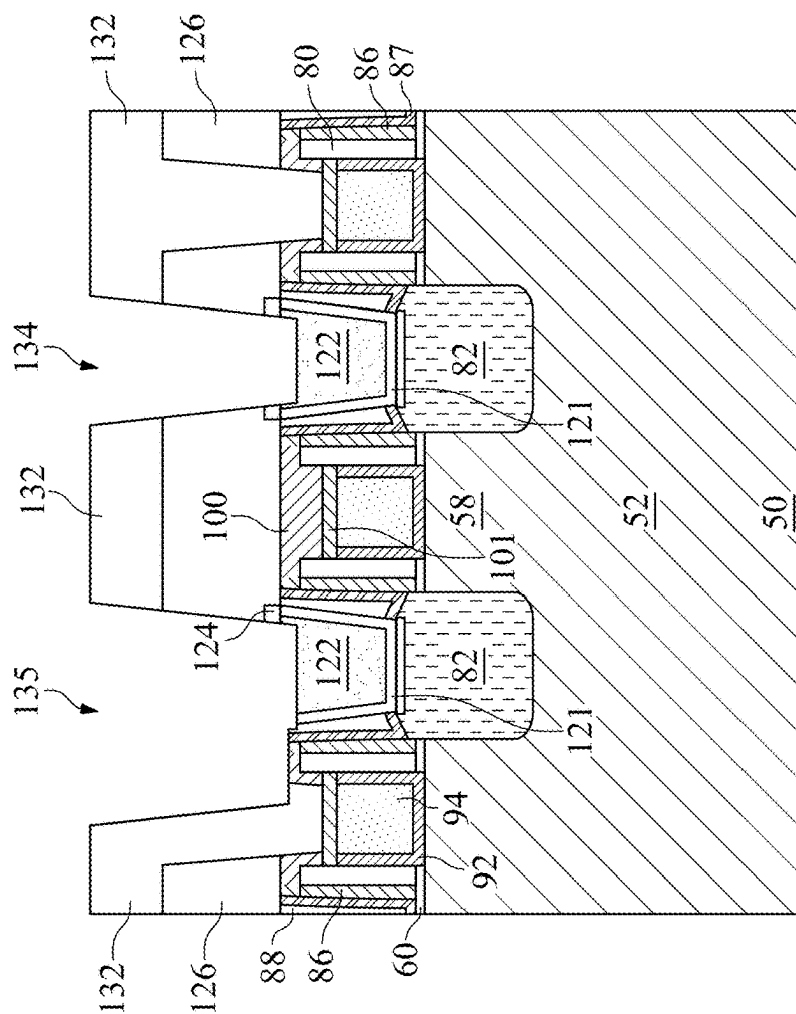


Fig. 22B

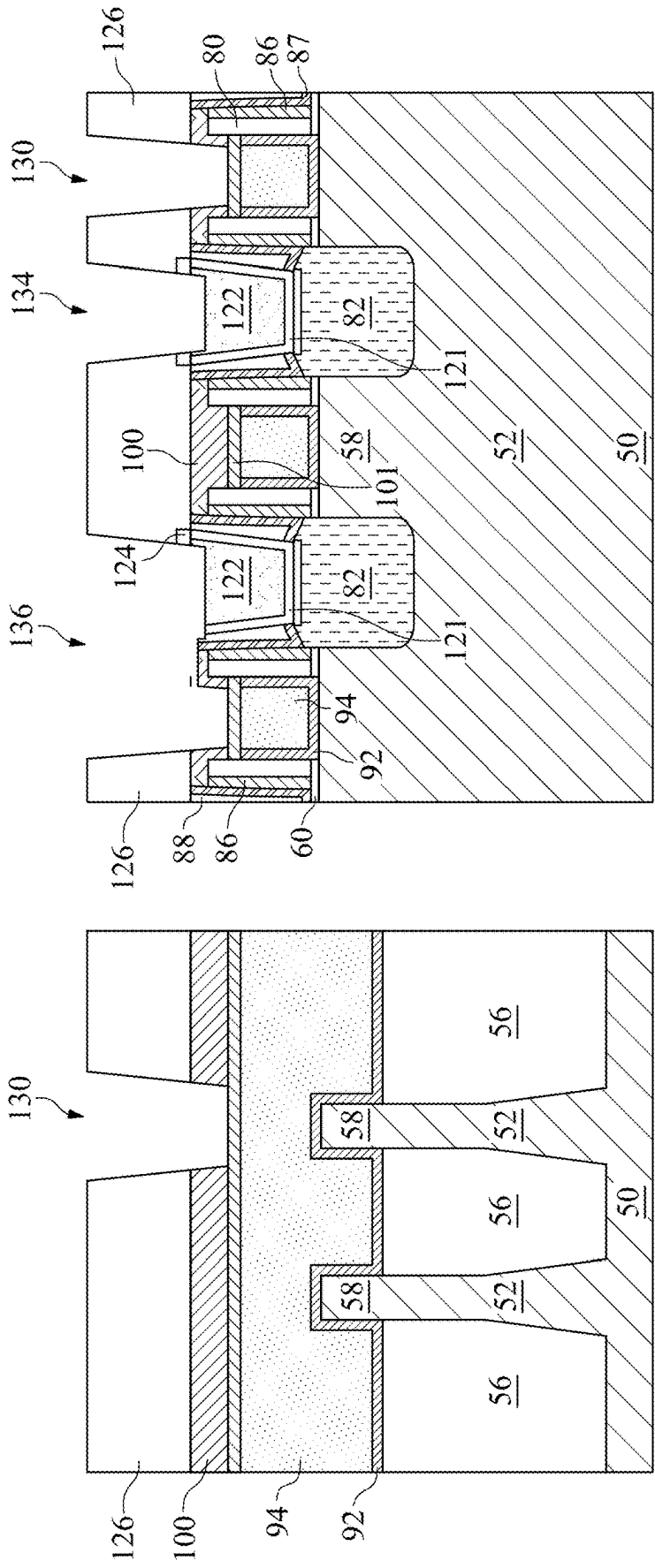


Fig. 23A

Fig. 23B

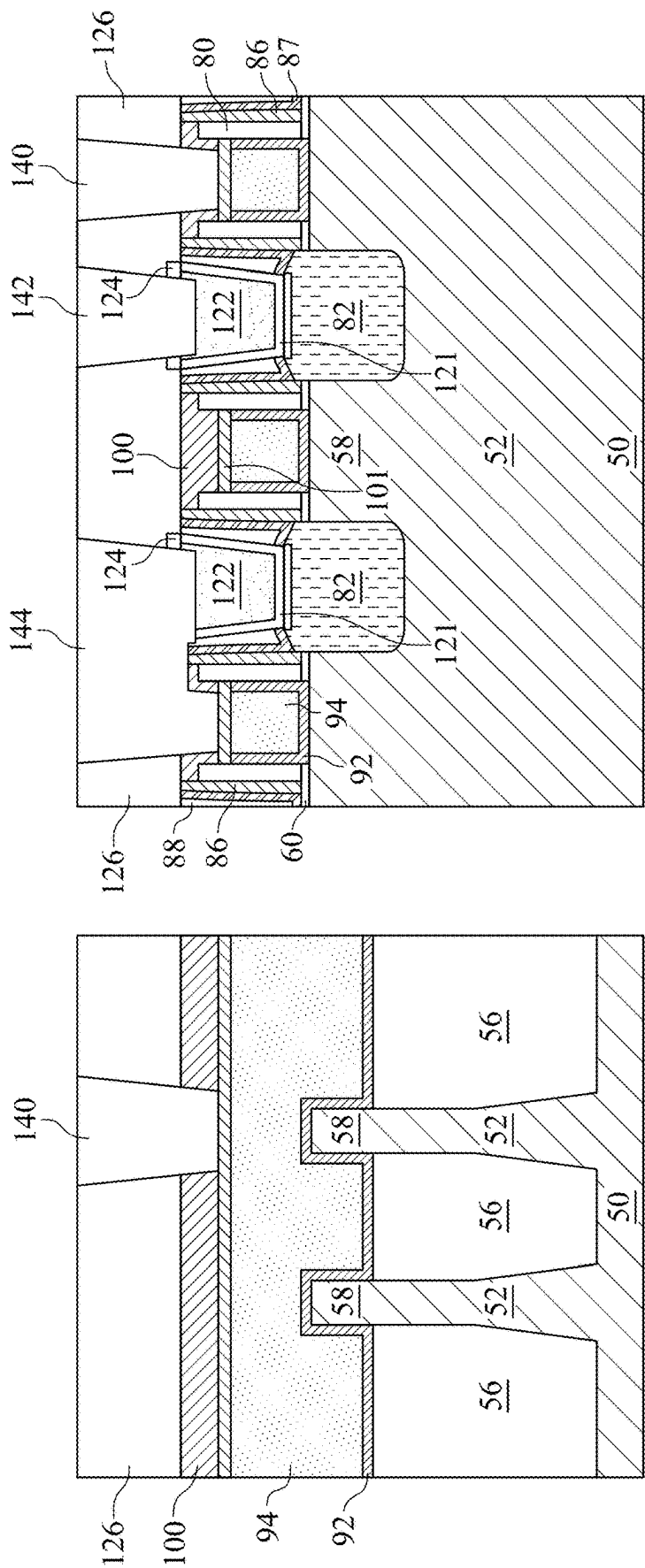


Fig. 24B

Fig. 24A

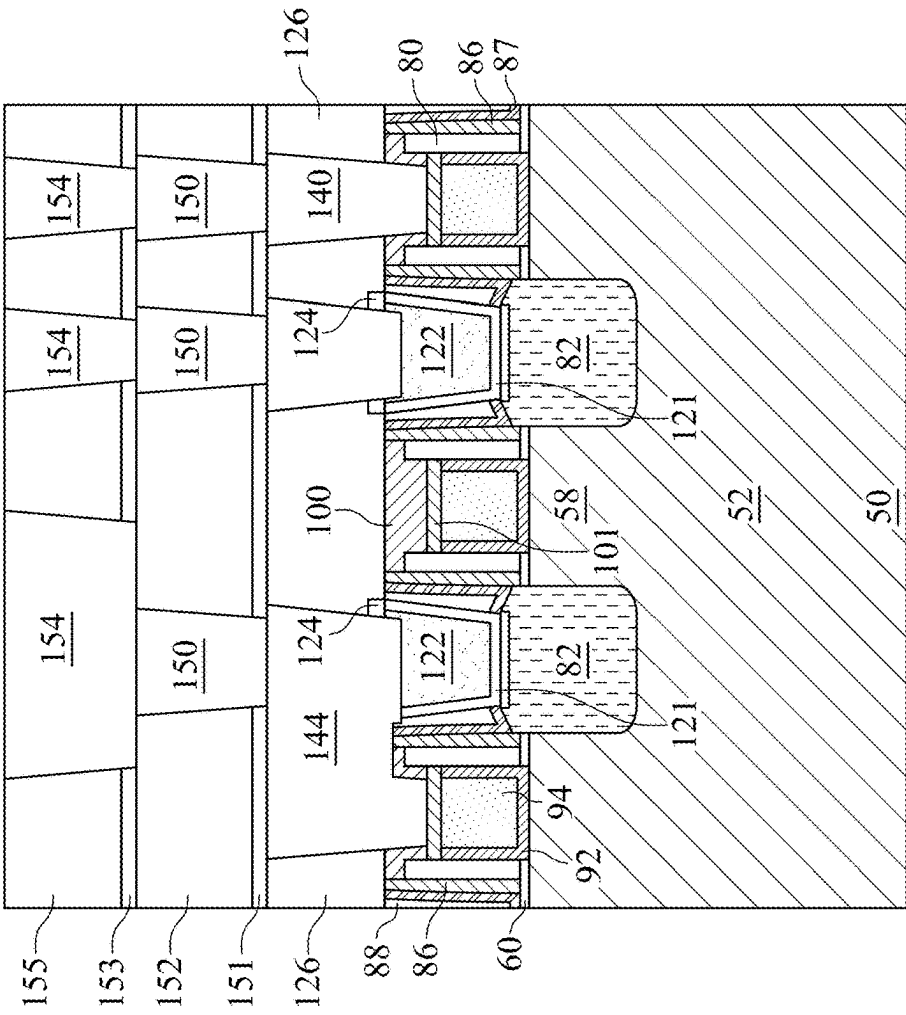


Fig. 25B

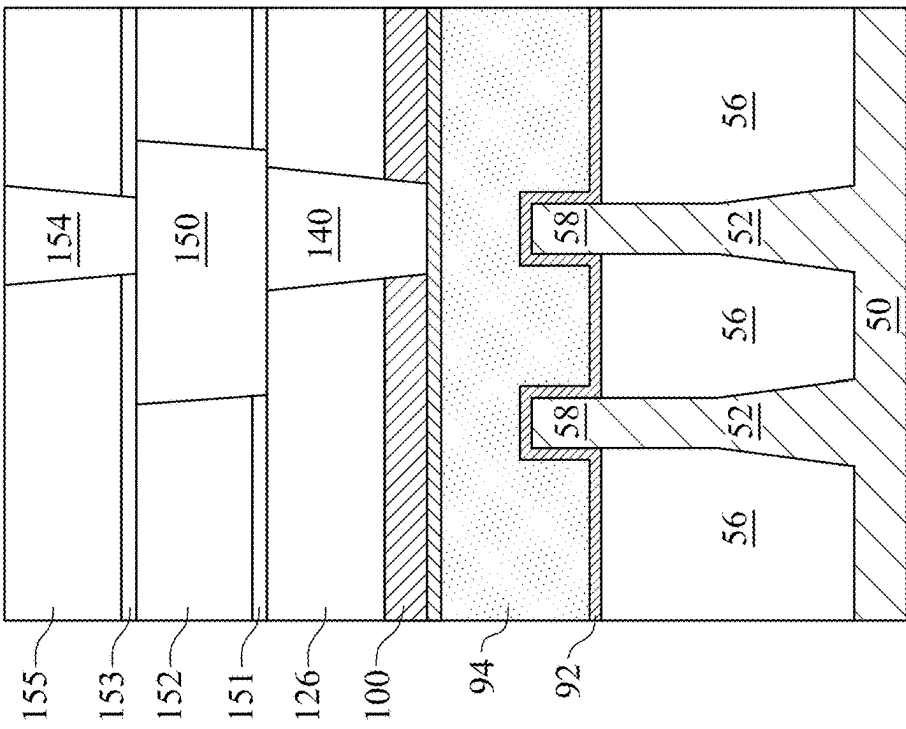


Fig. 25A

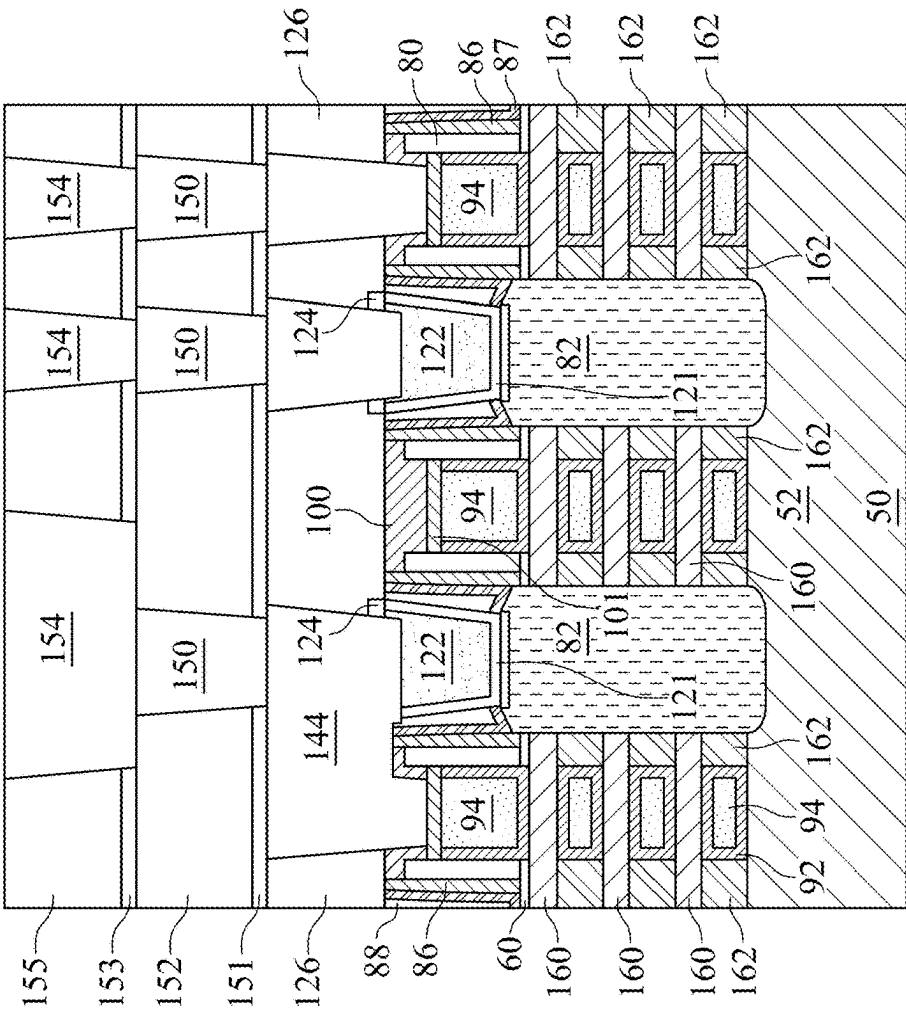


Fig. 26B

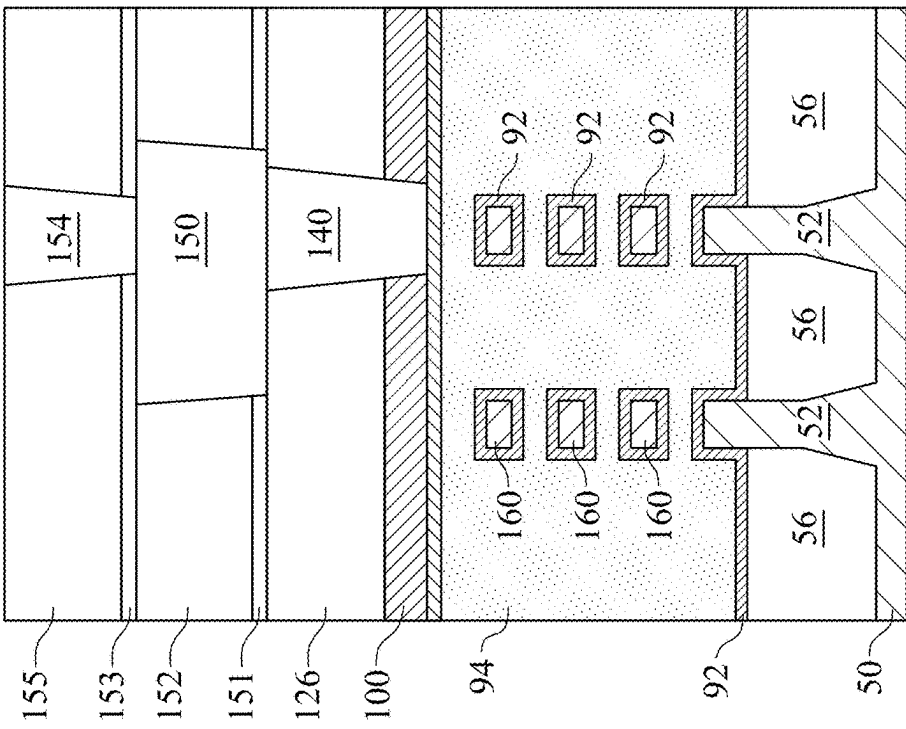


Fig. 26A

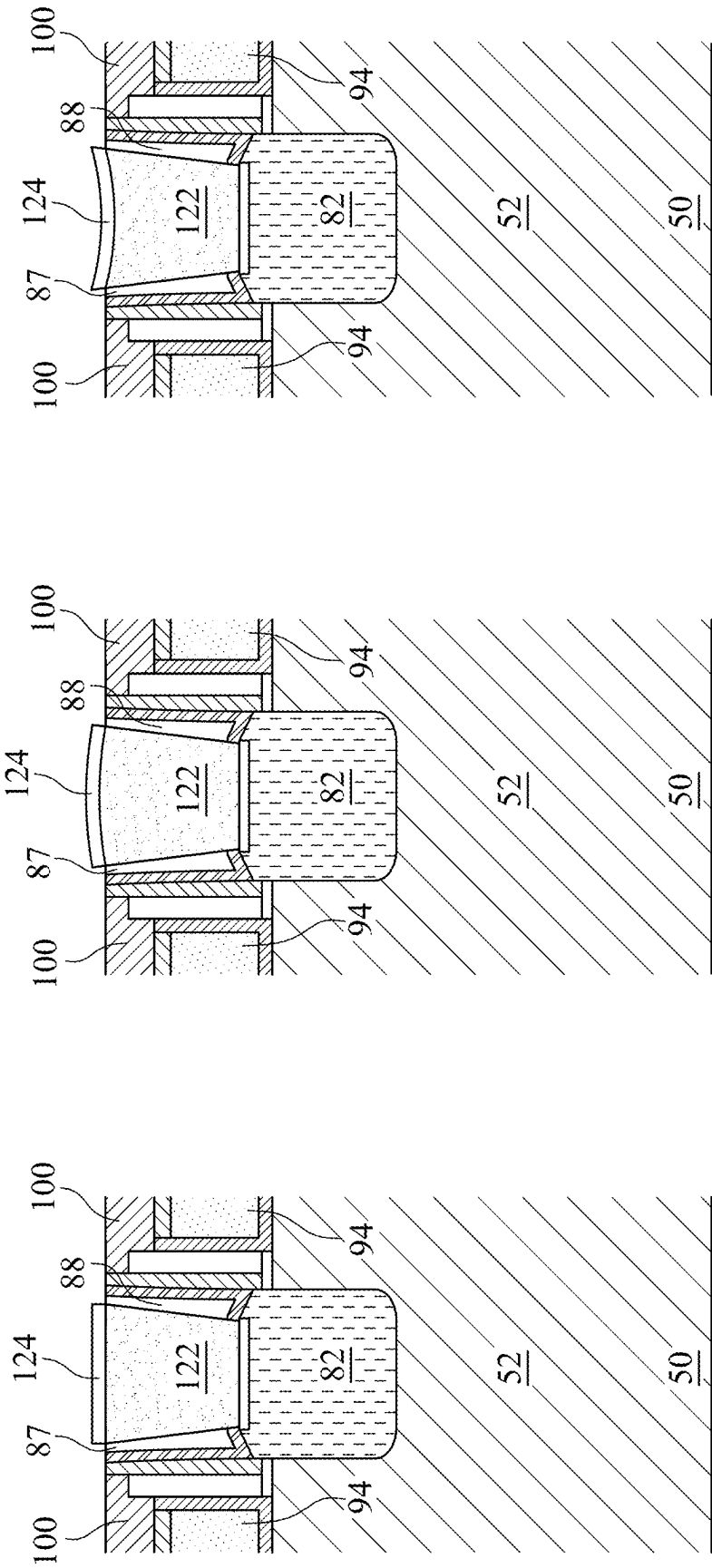


Fig. 27A

Fig. 27B

Fig. 27C

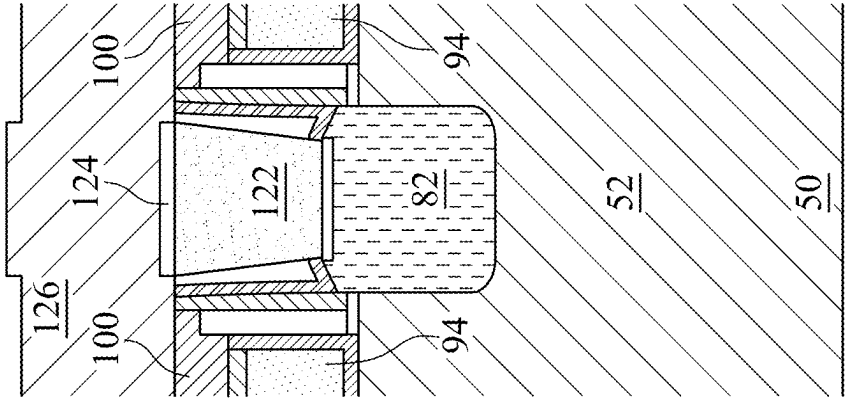


Fig. 28A

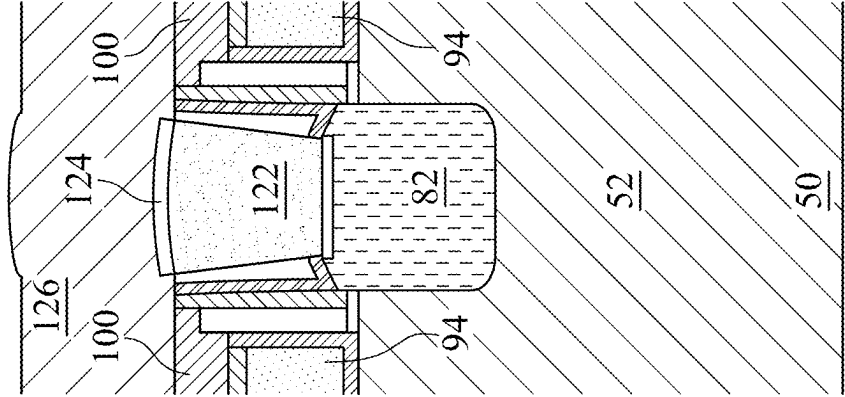


Fig. 28B

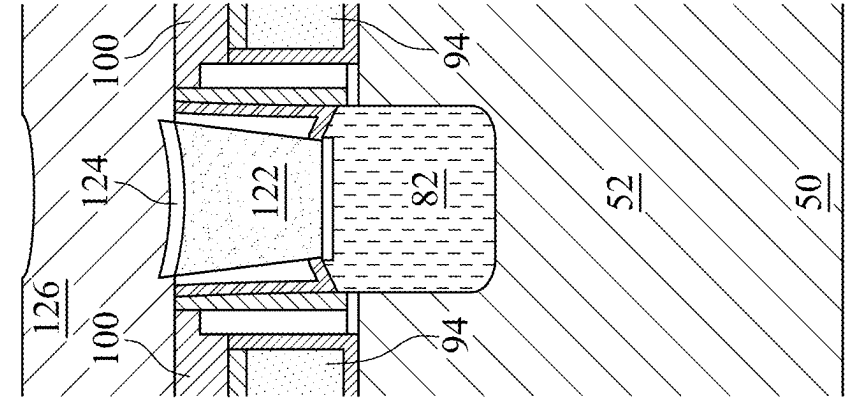


Fig. 28C

SEMICONDUCTOR DEVICES AND METHODS

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] This application is a continuation of U.S. patent application Ser. No. 17/831,884, filed on Jun. 3, 2022, entitled “Semiconductor Devices and Methods,” which claims the benefit of U.S. Provisional Application No. 63/268,182, filed on Feb. 17, 2022, which applications are hereby incorporated herein by reference.

BACKGROUND

[0002] Semiconductor devices are used in a variety of electronic applications, such as, for example, personal computers, cell phones, digital cameras, and other electronic equipment. Semiconductor devices are typically fabricated by sequentially depositing insulating or dielectric layers, conductive layers, and semiconductor layers of material over a semiconductor substrate, and patterning the various material layers using lithography to form circuit components and elements thereon.

[0003] The semiconductor industry continues to improve the integration density of various electronic components (e.g., transistors, diodes, resistors, capacitors, etc.) by continual reductions in minimum feature size, which allow more components to be integrated into a given area.

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0005] FIG. 1 illustrates an example of a Fin Field-Effect Transistor (FinFET) in a three-dimensional view, in accordance with some embodiments.

[0006] FIGS. 2, 3, 4, 5, 6, 7, 8A, 8B, 9A, 9B, 10A, 10B, 10C, 10D, 11A, 11B, 12A, 12B, 13A, 13B, 14A, 14B, 14C, 15A, 15B, 16A, 16B, 17A, 17B, 18A, 18B, 19A, 19B, 20A, 20B, 21A, 21B, 22A, 22B, 23A, 23B, 24A, 24B, 25A, 25B, 27A, 27B, 27C, 28A, 28B, and 28C are cross-sectional views of intermediate stages in the manufacturing of FinFET devices, in accordance with some embodiments.

[0007] FIGS. 26A and 26B are cross-sectional views of intermediate stages in the manufacturing of Nanostructure Field-Effect Transistor (NSFET) devices, in accordance with some embodiments.

DETAILED DESCRIPTION

[0008] The following disclosure provides many different embodiments, or examples, for implementing different features of the invention. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In

addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0009] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0010] Embodiments will be described with respect to a specific context, namely, a contact plug structure of a semiconductor device and a method of forming the same. Various embodiments presented herein are discussed in the context of a Fin Field Effect Transistor (FinFET) device formed using a gate-last process. In other embodiments, a gate-first process may be used. Various embodiments may be applied, however, to devices comprising other types of transistors such as planar FETs, nanostructure (e.g., nanosheet, nanowire, gate-all-around (GAA), or the like) field effect transistors (NFETs/NSFETs), or the like in lieu of or in combination with the FinFETs. In some embodiments, etch stop layers are selectively formed on contact plugs of a semiconductor device and used during subsequent processing steps, such as forming conductive features on the contact plugs. By forming the etch stop layers on the contact plugs, the overall thickness of the device can be reduced which leads to better profiles of conductive features over the gate stacks, source/drain regions, or the like, thereby improving electrical connections in the device.

[0011] FIG. 1 illustrates an example of a FinFET in a three-dimensional view, in accordance with some embodiments. The FinFET comprises a fin 52 on a substrate 50 (e.g., a semiconductor substrate). Isolation regions 56 are disposed in the substrate 50, and the fin 52 protrudes above and from between neighboring isolation regions 56. Although the isolation regions 56 are described/illustrated as being separate from the substrate 50, as used herein the term “substrate” may be used to refer to just the semiconductor substrate or a semiconductor substrate inclusive of isolation regions. Additionally, although the fin 52 is illustrated as a single, continuous material as the substrate 50, the fin 52 and/or the substrate 50 may comprise a single material or a plurality of materials. In this context, the fin 52 refers to the portion extending between the neighboring isolation regions 56.

[0012] A gate dielectric layer 92 is along sidewalls and over a top surface of the fin 52, and a gate electrode 94 is over the gate dielectric layer 92. The epitaxial source/drain regions 82 are disposed in opposite sides of the fin 52 with respect to the gate dielectric layer 92 and the gate electrode 94. FIG. 1 further illustrates reference cross-sections that are used in later figures. Cross-section A-A is along a longitudinal axis of the gate electrode 94 and in a direction, for example, perpendicular to the direction of current flow between the epitaxial source/drain regions 82 of the FinFET. Cross-section B-B is perpendicular to cross-section A-A and is along a longitudinal axis of the fin 52 and in a direction

of, for example, a current flow between the epitaxial source/drain regions **82** of the FinFET. Cross-section C-C is parallel to cross-section A-A and extends through a source/drain region of the FinFET. Subsequent figures refer to these reference cross-sections for clarity.

[0013] FIGS. **2** through **25B** are cross-sectional views of intermediate stages in the manufacturing of FinFET devices, in accordance with some embodiments. FIGS. **2**, **3**, **4**, **5**, **6**, **7**, **8A**, **9A**, **10A**, **11A**, **12A**, **13A**, **14A**, **15A**, **16A**, **17A**, **18A**, **19A**, **20A**, **21A**, **22A**, **23A**, **24A**, **25A** are illustrated along reference cross-section A-A illustrated in FIG. **1**, wherein the cross-sectional views illustrate multiple fins/FinFETs for illustrative purposes. FIGS. **8B**, **9B**, **10B**, **11B**, **12B**, **13B**, **14B**, **14C**, **15B**, **16B**, **17B**, **18B**, **19B**, **20B**, **21B**, **22B**, **23B**, **24B**, and **25B** are illustrated along reference cross-section B-B illustrated in FIG. **1**, wherein the cross-sectional views illustrate multiple fins/FinFETs for illustrative purposes. FIGS. **10C** and **10D** are illustrated along reference cross-section C-C illustrated in FIG. **1**, wherein the cross-sectional views illustrate multiple fins/FinFETs for illustrative purposes.

[0014] In FIG. **2**, a substrate **50** is provided. The substrate **50** may be a semiconductor substrate, such as a bulk semiconductor, a semiconductor-on-insulator (SOI) substrate, or the like, which may be doped (e.g., with a p-type or an n-type dopant) or undoped. The substrate **50** may be a wafer, such as a silicon wafer. Generally, an SOI substrate is a layer of a semiconductor material formed on an insulator layer. The insulator layer may be, for example, a buried oxide (BOX) layer, a silicon oxide layer, or the like. The insulator layer is provided on a substrate, typically a silicon or glass substrate. Other substrates, such as a multi-layered or gradient substrate may also be used. In some embodiments, the semiconductor material of the substrate **50** may include silicon; germanium; a compound semiconductor including silicon carbide, gallium arsenide, gallium phosphide, indium phosphide, indium arsenide, and/or indium antimonide; an alloy semiconductor including silicon germanium, gallium arsenide phosphide, aluminum indium arsenide, aluminum gallium arsenide, gallium indium arsenide, gallium indium phosphide, and/or gallium indium arsenide phosphide; the like; or combinations thereof.

[0015] The substrate **50** has an n-type region **50N** and a p-type region **50P**. The n-type region **50N** can be for forming n-type devices, such as NMOS transistors, e.g., n-type FinFETs. The p-type region **50P** can be for forming p-type devices, such as PMOS transistors, e.g., p-type FinFETs. The n-type region **50N** may be physically separated from the p-type region **50P** (as illustrated by divider **51**), and any number of device features (e.g., other active devices, doped regions, isolation structures, etc.) may be disposed between the n-type region **50N** and the p-type region **50P**.

[0016] In FIG. **3**, fins **52** are formed in the substrate **50**, in accordance with some embodiments. The fins **52** are semiconductor strips. In some embodiments, the fins **52** may be formed in the substrate **50** by etching trenches in the substrate **50**. The etching may be any acceptable etch process, such as a reactive ion etching (RIE), neutral beam etching (NBE), the like, or a combination thereof. The etching may be anisotropic.

[0017] The fins **52** may be patterned by any suitable method. For example, the fins **52** may be patterned using one or more photolithography processes, including double-patterning or multi-patterning processes. Generally, double-

patterning or multi-patterning processes combine photolithography and self-aligned processes, allowing patterns to be created that have, for example, pitches smaller than what is otherwise obtainable using a single, direct photolithography process. For example, in one embodiment, a sacrificial layer is formed over a substrate and patterned using a photolithography process. Spacers are formed alongside the patterned sacrificial layer using a self-aligned process. The sacrificial layer is then removed, and the remaining spacers may then be used as a mask to form the fins **52**. In some embodiments, the mask (or other layer) may remain on the fins **52**.

[0018] In FIG. **4**, an insulation material **54** is formed over the substrate **50** and between neighboring fins **52**, in accordance with some embodiments. The insulation material **54** may be an oxide, such as silicon oxide, a nitride, the like, or a combination thereof, and may be formed by a high-density plasma chemical vapor deposition (HDP-CVD), a flowable CVD (FCVD) (e.g., a CVD-based material deposition in a remote plasma system and post curing to make it convert to another material, such as an oxide), the like, or a combination thereof. Other insulation materials formed by any acceptable process may be used. In the illustrated embodiment, the insulation material **54** is silicon oxide formed by a FCVD process. An annealing process may be performed once the insulation material is formed. In an embodiment, the insulation material **54** is formed such that excess insulation material **54** covers the fins **52**. Although the insulation material **54** is illustrated as a single layer, some embodiments may utilize multiple layers. For example, in some embodiments a liner (not shown) may first be formed along surfaces of the substrate **50** and the fins **52**. Thereafter, a fill material such as those discussed above may be formed over the liner.

[0019] In FIG. **5**, a removal process is applied to the insulation material **54** to remove excess insulation material **54** over the fins **52**. In some embodiments, a planarization process such as a chemical mechanical polish (CMP), an etch-back process, a combination thereof, or the like may be utilized. The planarization process exposes the fins **52** such that top surfaces of the fins **52** and the insulation material **54** are substantially coplanar or level (e.g., within process variations of the planarization process) after the planarization process is completed. In embodiments in which a mask remains on the fins **52**, the planarization process may expose the mask or remove the mask such that top surfaces of the mask or the fins **52**, respectively, and the insulation material **54** are level after the planarization process is completed.

[0020] In FIG. **6**, the insulation material **54** is recessed to form Shallow Trench Isolation (STI) regions **56**, in accordance with some embodiments. The insulation material **54** is recessed such that upper portions of fins **52** in the n-type region **50N** and in the p-type region **50P** protrude from between neighboring STI regions **56**. Further, the top surfaces of the STI regions **56** may have a flat surface as illustrated, a convex surface, a concave surface (such as dishing), or a combination thereof. The top surfaces of the STI regions **56** may be formed flat, convex, and/or concave by an appropriate etch. The STI regions **56** may be recessed using an acceptable etch process, such as one that is selective to the material of the insulation material **54** (e.g., etches the material of the insulation material **54** at a faster rate than the material of the fins **52**). For example, an oxide removal

process using dilute hydrofluoric acid (dHF) may be used, though other processes are possible.

[0021] The process described with respect to FIGS. 2 through 6 is just one example of how the fins 52 may be formed. In some embodiments, the fins may be formed by an epitaxial growth process. For example, a dielectric layer can be formed over a top surface of the substrate 50, and trenches can be etched through the dielectric layer to expose the underlying substrate 50. Homoepitaxial structures can be epitaxially grown in the trenches, and the dielectric layer can be recessed such that the homoepitaxial structures protrude from the dielectric layer to form fins. Additionally, in some embodiments, heteroepitaxial structures can be used for the fins 52. For example, the fins 52 in FIG. 5 can be recessed, and a material different from the fins 52 may be epitaxially grown over the recessed fins 52. In such embodiments, the fins 52 comprise the recessed material as well as the epitaxially grown material disposed over the recessed material. In an even further embodiment, a dielectric layer can be formed over a top surface of the substrate 50, and trenches can be etched through the dielectric layer. Heteroepitaxial structures can then be epitaxially grown in the trenches using a material different from the substrate 50, and the dielectric layer can be recessed such that the heteroepitaxial structures protrude from the dielectric layer to form the fins 52. In some embodiments where homoepitaxial or heteroepitaxial structures are epitaxially grown, the epitaxially grown materials may be in situ doped during growth, which may obviate prior and subsequent implantations although in situ and implantation doping may be used together.

[0022] Still further, it may be advantageous to epitaxially grow a material in n-type region 50N (e.g., an NMOS region) different from the material in p-type region 50P (e.g., a PMOS region). In various embodiments, upper portions of the fins 52 may be formed from silicon germanium ($\text{Si}_x\text{Ge}_{1-x}$, where x can be in the range of 0 to 1), silicon carbide, pure or substantially pure germanium, a III-V compound semiconductor, a II-VI compound semiconductor, or the like. For example, the available materials for forming III-V compound semiconductor include, but are not limited to, indium arsenide, aluminum arsenide, gallium arsenide, indium phosphide, gallium nitride, indium gallium arsenide, indium aluminum arsenide, gallium antimonide, aluminum antimonide, aluminum phosphide, gallium phosphide, and the like.

[0023] Further in FIG. 6, appropriate wells (not shown) may be formed in the fins 52 and/or the substrate 50. In some embodiments, a P well may be formed in the n-type region 50N, and an N well may be formed in the p-type region 50P. In some embodiments, a P well or an N well are formed in both the n-type region 50N and the p-type region 50P. In the embodiments with different well types, the different implantation steps for the n-type region 50N and the p-type region 50P may be achieved using a photoresist and/or other masks (not shown). For example, a photoresist may be formed over the fins 52 and the STI regions 56 in the n-type region 50N. The photoresist is patterned to expose the p-type region 50P of the substrate 50. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, an n-type impurity implantation is performed in the p-type region 50P, and the photoresist may act as a mask to substantially prevent n-type impurities from being implanted into the n-type region 50N. The n-type impurities

may be phosphorus, arsenic, antimony, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as in the range of about 10^{16} cm^{-3} to about 10^{18} cm^{-3} . After the implantation, the photoresist is removed, such as by an acceptable ashing process.

[0024] Following the implanting of the p-type region 50P, a photoresist is formed over the fins 52 and the STI regions 56 in the p-type region 50P. The photoresist is patterned to expose the n-type region 50N of the substrate 50. The photoresist can be formed by using a spin-on technique and can be patterned using acceptable photolithography techniques. Once the photoresist is patterned, a p-type impurity implantation may be performed in the n-type region 50N, and the photoresist may act as a mask to substantially prevent p-type impurities from being implanted into the p-type region 50P. The p-type impurities may be boron, boron fluoride, indium, or the like implanted in the region to a concentration of equal to or less than 10^{18} cm^{-3} , such as in the range of about 10^{16} cm^{-3} to about 10^{18} cm^{-3} . After the implantation, the photoresist may be removed, such as by an acceptable ashing process.

[0025] After the implanting of the n-type region 50N and the p-type region 50P, an annealing may be performed to repair implantation damage and to activate the p-type and/or n-type impurities that were implanted. In some embodiments, the grown materials of epitaxial fins may be in situ doped during growth, which may obviate the implantations, although in situ and implantation doping may be used together.

[0026] In FIG. 7, a dummy dielectric layer 60 is formed on the fins 52. The dummy dielectric layer 60 may be, for example, silicon oxide, silicon nitride, a combination thereof, or the like, and may be deposited or thermally grown according to acceptable techniques. A dummy gate layer 62 is formed over the dummy dielectric layer 60, and a mask layer 64 is formed over the dummy gate layer 62. The dummy gate layer 62 may be deposited over the dummy dielectric layer 60 and then planarized using, for example, a CMP process. The mask layer 64 may be deposited over the dummy gate layer 62. The dummy gate layer 62 may be a conductive material and may be selected from a group including amorphous silicon, polycrystalline-silicon (poly-silicon), poly-crystalline silicon-germanium (poly-SiGe), metallic nitrides, metallic silicides, metallic oxides, and metals. The dummy gate layer 62 may be deposited by physical vapor deposition (PVD), CVD, sputter deposition, or other techniques known and used in the art for depositing conductive materials. The dummy gate layer 62 may be made of other materials that have a high etching selectivity than materials of the STI regions 56. The mask layer 64 may include, for example, one or more layers of silicon oxide, SiN, SiON, a combination thereof, or the like. In some embodiments, the mask layer 64 may comprise a layer of silicon nitride and a layer of silicon oxide over the layer of silicon nitride. In some embodiments, a single dummy gate layer 62 and a single mask layer 64 are formed across the region 50N and the region 50P. It is noted that the dummy dielectric layer 60 is shown covering only the fins 52 for illustrative purposes only. In some embodiments, the dummy dielectric layer 60 may be deposited such that the dummy dielectric layer 60 covers the STI regions 56, extending between the dummy gate layer 62 and the STI regions 56.

[0027] FIGS. 8A through 26B illustrate various additional steps in the manufacturing of embodiment devices. FIGS. 8A through 26B illustrate features in either of the n-type region 50N and the p-type region 50P. For example, the structures illustrated in FIGS. 8A through 26B may be applicable to both the n-type region 50N and the p-type region 50P. Differences (if any) in the structures of the n-type region 50N and the p-type region 50P are described in the text accompanying each figure.

[0028] In FIGS. 8A and 8B, the mask layer 64 (see FIG. 7) may be patterned using acceptable photolithography and etching techniques to form masks 74. The pattern of the masks 74 then may be transferred to the dummy gate layer 62. In some embodiments (not illustrated), the pattern of the masks 74 may also be transferred to the dummy dielectric layer 60 by an acceptable etching technique to form dummy gates 72. The dummy gates 72 cover respective channel regions 58 of the fins 52. The pattern of the masks 74 may be used to physically separate each of the dummy gates 72 from adjacent dummy gates. The dummy gates 72 may also have a lengthwise direction substantially perpendicular to the lengthwise direction of respective epitaxial fins 52.

[0029] Further in FIGS. 8A and 8B, gate seal spacers 80 can be formed on exposed surfaces of the dummy gates 72, the masks 74, and/or the fins 52. A thermal oxidation or a deposition followed by an anisotropic etch may form the gate seal spacers 80. The gate seal spacers 80 may be formed of silicon oxide, silicon nitride, silicon oxynitride, or the like.

[0030] After the formation of the gate seal spacers 80, implantations for lightly doped source/drain (LDD) regions (not explicitly illustrated) may be performed. In the embodiments with different device types, similar to the implantations discussed above in FIG. 6, a mask, such as a photoresist, may be formed over the n-type region 50N, while exposing the p-type region 50P, and appropriate type (e.g., p-type) impurities may be implanted into the exposed fins 52 in the p-type region 50P. The mask may then be removed. Subsequently, a mask, such as a photoresist, may be formed over the p-type region 50P while exposing the n-type region 50N, and appropriate type (e.g., n-type) impurities may be implanted into the exposed fins 52 in the n-type region 50N. The mask may then be removed. The n-type impurities may be any of the n-type impurities previously discussed, and the p-type impurities may be any of the p-type impurities previously discussed. In some embodiments, the lightly doped source/drain regions may have a concentration of impurities in the range of about 10^{15} cm^{-3} to about 10^{19} cm^{-3} . An annealing may be used to repair implantation damage and/or to activate the implanted impurities.

[0031] In FIGS. 9A and 9B, gate spacers 86 are formed on the gate seal spacers 80 along sidewalls of the dummy gates 72 and the masks 74. The gate spacers 86 may be formed by conformally depositing an insulating material and subsequently anisotropically etching the insulating material. The insulating material of the gate spacers 86 may be silicon oxide, silicon nitride, silicon oxynitride, silicon carbonitride, a combination thereof, or the like. In some embodiments, the gate spacers 86 comprise multiple layers, which may be layers of different materials.

[0032] It is noted that the above disclosure generally describes a process of forming spacers and LDD regions. Other processes and sequences may be used. For example, fewer or additional spacers may be utilized, and different

sequences of steps may be utilized (e.g., the gate seal spacers 80 may not be etched prior to forming the gate spacers 86, yielding “L-shaped” gate seal spacers, spacers may be formed and removed, and/or the like). Furthermore, the n-type and p-type devices may be formed using a different structures and steps. For example, LDD regions for n-type devices may be formed prior to forming the gate seal spacers 80 while the LDD regions for p-type devices may be formed after forming the gate seal spacers 80.

[0033] In FIGS. 10A and 10B epitaxial source/drain regions 82 are formed in the fins 52. The epitaxial source/drain regions 82 are formed in the fins 52 such that each dummy gate 72 is disposed between respective neighboring pairs of the epitaxial source/drain regions 82. In some embodiments the epitaxial source/drain regions 82 may extend into, and may also penetrate through, the fins 52. In some embodiments, the gate spacers 86 are used to separate the epitaxial source/drain regions 82 from the dummy gates 72 by an appropriate lateral distance so that the epitaxial source/drain regions 82 do not short out subsequently formed gates of the resulting FinFETs. A material of the epitaxial source/drain regions 82 may be selected to exert stress in the respective channel regions 58, thereby improving performance.

[0034] The epitaxial source/drain regions 82 in the n-type region 50N may be formed by masking the p-type region 50P and etching source/drain regions of the fins 52 in the n-type region 50N to form recesses in the fins 52. Then, the epitaxial source/drain regions 82 in the n-type region 50N are epitaxially grown in the recesses. The epitaxial source/drain regions 82 may include any acceptable material, such as appropriate for n-type FinFETs. For example, if the fin 52 is silicon, the epitaxial source/drain regions 82 in the n-type region 50N may include materials exerting a tensile strain in the channel region 58, such as silicon, silicon carbide, phosphorous doped silicon carbide, silicon phosphide, or the like. The epitaxial source/drain regions 82 in the n-type region 50N may have surfaces raised from respective surfaces of the fins 52 and may have facets.

[0035] The epitaxial source/drain regions 82 in the p-type region 50P may be formed by masking the n-type region 50N and etching source/drain regions of the fins 52 in the p-type region 50P to form recesses in the fins 52. Then, the epitaxial source/drain regions 82 in the p-type region 50P are epitaxially grown in the recesses. The epitaxial source/drain regions 82 may include any acceptable material, such as appropriate for p-type FinFETs. For example, if the fin 52 is silicon, the epitaxial source/drain regions 82 in the p-type region 50P may comprise materials exerting a compressive strain in the channel region 58, such as silicon-germanium, boron doped silicon-germanium, germanium, germanium tin, or the like. The epitaxial source/drain regions 82 in the p-type region 50P may have surfaces raised from respective surfaces of the fins 52 and may have facets.

[0036] The epitaxial source/drain regions 82 and/or the fins 52 may be implanted with dopants to form source/drain regions, similar to the process previously discussed for forming lightly-doped source/drain regions, followed by an annealing. The source/drain regions may have an impurity concentration in the range of about 10^{19} cm^{-3} to about 10^{21} cm^{-3} . The n-type and/or p-type impurities for source/drain regions may be any of the impurities previously discussed. In some embodiments, the epitaxial source/drain regions 82 may be in situ doped during growth.

[0037] As a result of the epitaxy processes used to form the epitaxial source/drain regions **82** in the n-type region **50N** and the p-type region **50P**, upper surfaces of the epitaxial source/drain regions have facets which expand laterally outward beyond sidewalls of the fins **52**. In some embodiments, these facets cause adjacent epitaxial source/drain regions **82** of a same FinFET to merge as illustrated by FIG. **10C**. In other embodiments, adjacent epitaxial source/drain regions **82** remain separated after the epitaxy process is completed as illustrated by FIG. **10D**. In the embodiments illustrated in FIGS. **10C** and **10D**, gate spacers **86** are formed covering a portion of the sidewalls of the fins **52** that extend above the STI regions **56**, thereby blocking the epitaxial growth. In some other embodiments, the spacer etch used to form the gate spacers **86** may be adjusted to remove the spacer material to allow the epitaxially grown region to extend to the surface of the STI region **56**.

[0038] In FIGS. **11A** and **11B**, a first interlayer dielectric (ILD) **88** is deposited over the structure illustrated in FIGS. **10A** and **10B**. The first ILD **88** may be formed of a dielectric material, and may be deposited by any suitable method, such as CVD, plasma-enhanced CVD (PECVD), or FCVD. Dielectric materials may include an oxide, phosphosilicate glass (PSG), borosilicate glass (BSG), boron-doped phosphosilicate glass (BPSG), undoped silicate glass (USG), or the like. Other insulation materials formed by any acceptable process may be used. In some embodiments, a contact etch stop layer (CESL) **87** is disposed between the first ILD **88** and the epitaxial source/drain regions **82**, the masks **74**, and the gate spacers **86**. The CESL **87** may comprise a dielectric material, such as, silicon nitride, silicon oxide, silicon oxynitride, or the like, having a lower etch rate than the material of the overlying first ILD **88**.

[0039] In FIGS. **12A** and **12B**, a planarization process, such as a CMP, may be performed to level the top surface of the first ILD **88** with the top surfaces of the dummy gates **72** or the masks **74**. The planarization process may also remove the masks **74** on the dummy gates **72**, and portions of the gate seal spacers **80** and the gate spacers **86** along sidewalls of the masks **74**. After the planarization process, top surfaces of the dummy gates **72**, the gate seal spacers **80**, the gate spacers **86**, and the first ILD **88** are level. Accordingly, the top surfaces of the dummy gates **72** are exposed through the first ILD **88**. In some embodiments, the masks **74** may remain, in which case the planarization process levels the top surface of the first ILD **88** with the top surfaces of the masks **74**.

[0040] In FIGS. **13A** and **13B**, the dummy gates **72**, and the masks **74** if present, are removed in an etching step(s), so that recesses **90** are formed. Portions of the dummy dielectric layer **60** in the recesses **90** may also be removed. In some embodiments, only the dummy gates **72** are removed and the dummy dielectric layer **60** remains and is exposed by the recesses **90**. In some embodiments, the dummy dielectric layer **60** is removed from recesses **90** in a first region of a die (e.g., a core logic region) and remains in recesses **90** in a second region of the die (e.g., an input/output region). In some embodiments, the dummy gates **72** are removed by an anisotropic dry etch process. For example, the etch process may include a dry etch process using reaction gas(es) that selectively etch the dummy gates **72** with little or no etching of the first ILD **88** or the gate spacers **86**. Each recess **90** exposes and/or overlies a channel region **58** of a respective fin **52**. Each channel region **58** is

disposed between neighboring pairs of the epitaxial source/drain regions **82**. During the removal, the dummy dielectric layer **60** may be used as an etch stop layer when the dummy gates **72** are etched. The dummy dielectric layer **60** may then be optionally removed after the removal of the dummy gates **72**.

[0041] In FIGS. **14A** and **14B**, gate dielectric layers **92** and gate electrodes **94** are formed for replacement gates. FIG. **14C** illustrates a detailed view of region **89** of FIG. **14B**. Gate dielectric layers **92** includes one or more layers deposited in the recesses **90**, such as on the top surfaces and the sidewalls of the fins **52** and on sidewalls of the gate seal spacers **80**/gate spacers **86**. The gate dielectric layers **92** may also be formed on the top surface of the first ILD **88**. In some embodiments, the gate dielectric layers **92** comprise one or more dielectric layers, such as one or more layers of silicon oxide, silicon nitride, metal oxide, metal silicate, or the like. For example, in some embodiments, the gate dielectric layers **92** include an interfacial layer of silicon oxide formed by thermal or chemical oxidation and an overlying high-k dielectric material, such as a metal oxide or a silicate of hafnium, aluminum, zirconium, lanthanum, manganese, barium, titanium, lead, and combinations thereof. The gate dielectric layers **92** may include a dielectric layer having a k-value greater than about 7.0. The formation methods of the gate dielectric layers **92** may include Molecular-Beam Deposition (MBD), atomic layer deposition (ALD), PECVD, and the like. In embodiments where portions of the dummy dielectric layer **60** remains in the recesses **90**, the gate dielectric layers **92** include a material of the dummy dielectric layer **60** (e.g., silicon oxide or the like).

[0042] The gate electrodes **94** are deposited over the gate dielectric layers **92**, respectively, and fill the remaining portions of the recesses **90**. The gate electrodes **94** may include a metal-containing material such as titanium nitride, titanium oxide, tantalum nitride, tantalum carbide, cobalt, ruthenium, aluminum, tungsten, combinations thereof, or multi-layers thereof. For example, although a single layer gate electrode **94** is illustrated in FIG. **14B**, the gate electrode **94** may comprise any number of liner layers **94A**, any number of work function tuning layers **94B**, and a fill material **94C** as illustrated by FIG. **14C**. After the filling of the recesses **90**, a planarization process, such as a CMP, may be performed to remove the excess portions of the gate dielectric layers **92** and the material of the gate electrodes **94**, which excess portions are over the top surface of the first ILD **88**. The remaining portions of material of the gate electrodes **94** and the gate dielectric layers **92** thus form replacement gates of the resulting FinFETs. The gate electrodes **94** and the gate dielectric layers **92** may be collectively referred to as a "replacement gate," a "gate structure," or a "gate stack." The gate and the gate stacks may extend along sidewalls of a channel region **58** of the fins **52**.

[0043] The formation of the gate dielectric layers **92** in the n-type region **50N** and the p-type region **50P** may occur simultaneously such that the gate dielectric layers **92** in each region are formed from the same materials, and the formation of the gate electrodes **94** may occur simultaneously such that the gate electrodes **94** in each region are formed from the same materials. In some embodiments, the gate dielectric layers **92** in each region may be formed by distinct processes, such that the gate dielectric layers **92** may be different materials, and/or the gate electrodes **94** in each region may be formed by distinct processes, such that the

gate electrodes **94** may be different materials. Various masking steps may be used to mask and expose appropriate regions when using distinct processes.

[0044] In FIGS. **15A** and **15B**, the gate stacks (e.g., the gate dielectric layers **92**, the gate electrodes **94**, the gate seal spacers **80**, and the gate spacers **86**) are recessed and gate contact layers **101** and capping layers **100** are formed over the gate stacks, in accordance with some embodiments. In an embodiment the gate dielectric layers **92**, the gate electrodes **94**, and, in some embodiments, the gate seal spacers **80** and the gate spacers **86** are recessed and gate contact layers **101** and capping layers **100** may be deposited over the recessed gate electrodes **94**. In an embodiment the gate dielectric layers **92**, the gate electrodes **94**, the gate seal spacers **80**, and the gate spacers **86** may be recessed using, e.g., one or more etching processes, such as wet etching processes or dry etching processes. The one or more etch processes may comprise anisotropic etch processes.

[0045] Additionally, because of the different materials involved, the recessing processes may cause one or more of the gate dielectric layers **92**, the gate electrodes **94**, the gate seal spacers **80**, and the gate spacers **86** to be recessed further than the other materials. For example, in a particular embodiment the gate electrodes **94** may be recessed further than the gate seal spacers **80** or the gate spacers **86**. Because of this, the gate electrodes **94** may have an upper surface facing away from the fins **52** that is closer to the fins **52** than an upper surface of either the gate seal spacers **80** or the gate spacers **86**, which helps to prevent short circuits caused by the gate electrodes **94** being too high. However, any suitable combination of recesses may be utilized.

[0046] Once the gate stack has been recessed, the gate contact layer **101** may be formed of tungsten, such as fluorine-free tungsten (FFW), which is deposited by a selective deposition process, such as a selective CVD process. However, the gate contact layer **101** may include other conductive materials, such as ruthenium, cobalt, copper, molybdenum, nickel, combinations thereof, or the like and may be deposited using a suitable deposition process (e.g., ALD, CVD, PVD, or the like).

[0047] The capping layers **100** may be a dielectric material that may be formed, for example, by depositing the dielectric material of the capping layers **100** on the gate contact layer **101** of the recessed gate stacks. In some embodiments, the capping layers **100** may comprise silicon nitride, silicon carbide, silicon carbonitride, another type of nitride, a combination thereof, or the like, and may be formed using ALD, CVD, PVD, a combination thereof, or the like. In some cases, using oxygen-free materials to form the capping layers **100** may reduce the oxidation on the gate electrodes **94**. In some embodiments, the capping layers **100** may comprise silicon oxide, silicon oxynitride, a metal oxide, another type of oxide, a combination thereof, or the like. The capping layers **100** may be formed in a self-aligned manner, and sidewalls of a capping layer **100** may be aligned with respective sidewalls of the gate seal spacers **80** or the gate spacers **86**. A planarization process, such as CMP process, may be performed to remove excess material of the capping layers **100** (e.g., from over the first ILD **88**). In some embodiments, top surfaces of the capping layers **100** and top surfaces of the first ILD **88** may be level. In some embodiments, the capping layers **100** may be formed having a thickness in the range of about 10 nm to about 20 nm.

[0048] FIGS. **16A** through **17B** illustrate the formation of conductive features **122** (see FIG. **17B**), in accordance with some embodiments. The conductive features **122** provide electrical connections to respective epitaxial source/drain regions **82** and in some cases may be considered “source/drain contact plugs” or the like.

[0049] FIGS. **16A** and **16B** illustrate a patterning process of the first ILD **88** and the CESL **87** to form openings **118**, in accordance with some embodiments. The openings **118** may expose surfaces of the epitaxial source/drain regions **82**. The patterning may be performed using acceptable photolithography and etching techniques. For example, a photoresist may be formed over the first ILD **88** and the capping layers **100** and patterned. The photoresist can be formed by using, for example, a spin-on technique and can be patterned using acceptable photolithography techniques. One or more suitable etch processes may be performed using the patterned photoresist as an etch mask, forming the openings **118**. The one or more etch processes may include wet and/or dry etch processes. One or more of the etch processes may be anisotropic. FIGS. **16A-16B** show the openings **118** as having sloped sidewalls, but the openings **118** may have substantially vertical sidewalls, curved sidewalls, or another sidewall profile than shown.

[0050] In FIGS. **17A** and **17B**, silicide layers **120** and conductive features **122** are formed in the openings **118**, in accordance with some embodiments. The silicide layers **120** may be formed, for example, by depositing a metallic material in the openings **118**. The metallic material may comprise Ti, Co, Ni, NiCo, Pt, NiPt, Ir, PtIr, Er, Yb, Pd, Rh, Nb, a combination thereof, or the like, and may be formed using ALD, CVD, PVD, sputtering, a combination thereof, or the like. Subsequently, an annealing process is performed to form the silicide layers **120**. In some embodiments in which the epitaxial source/drain regions **82** comprise silicon, the annealing process may cause the metallic material to react with silicon to form a silicide of the metallic material at interfaces between the metallic material and the epitaxial source/drain regions **82**. After forming the silicide layers **120**, unreacted portions of the metallic material may be removed using a suitable removal process, such as a suitable etch process, for example.

[0051] After forming the silicide layers **120**, conductive features **122** are formed in the openings **118**. The conductive features **122** provide electrical connections to respective epitaxial source/drain regions **82**. In some embodiments, the conductive features **122** are formed by initially forming a liner **121**, such as a barrier layer, an adhesion layer, or the like, and the conductive features **122** are formed to fill a remainder of the openings **118**. For example, a barrier layer may first be formed in the openings **118**. The barrier layer may extend along a bottom and sidewalls of the openings **118**. The barrier layer may comprise titanium, titanium nitride, tantalum, tantalum nitride, a combination thereof, a multilayer thereof, or the like, and may be formed by ALD, CVD, PVD, sputtering, a combination thereof, or the like. Subsequently, an adhesion layer (not individually shown) may be formed over the barrier layer within the openings **118**. The adhesion layer may comprise cobalt, ruthenium, an alloy thereof, a combination thereof, a multilayer thereof, or the like, and may be formed by ALD, CVD, PVD, sputtering, a combination thereof, or the like. The barrier layer and/or the adhesion layer may be omitted in other embodiments.

[0052] A conductive fill material is then formed in the openings 118 to form the conductive features 122. The conductive fill material may comprise cobalt, tungsten, ruthenium, copper, combinations thereof, alloys thereof, multilayers thereof, or the like, and may be formed using, for example, by plating, ALD, CVD, PVD, or other suitable methods. For example, in some embodiments, the conductive fill material may be formed by first forming a seed layer (not individually shown) over the adhesion layer within the openings 118. The seed layer may comprise copper, titanium, nickel, gold, manganese, a combination thereof, a multilayer thereof, or the like, and may be formed by ALD, CVD, PVD, sputtering, a combination thereof, or the like. The conductive fill material may then be formed over the seed layer within the openings 118. Other techniques for forming the conductive fill material are possible.

[0053] In some embodiments, the conductive fill material overfills the openings 118. After forming the conductive fill material, a planarization process may be performed to remove portions of the conductive fill material overfilling the openings 118. If present, portions of the barrier layer, the adhesion layer, and/or the seed layer may also be removed. Remaining portions of the barrier layer, the adhesion layer, the seed layer, and the conductive fill material form the conductive features 122 in the openings 118. The planarization process may comprise a CMP process, an etching back process, a grinding process, combinations thereof, or the like. After performing the planarization process, top surfaces of the conductive features 122 and the top surfaces of the capping layers 100 may be substantially level. In other embodiments, a planarization process is not performed. In some embodiments, an optional annealing process is performed after the planarization process to recrystallize the conductive features 122, to enlarge the grain structure of the conductive features 122, to reduce micro-voids in the conductive features 122, and/or to reduce impurities in the conductive features 122.

[0054] Of course, the above described process for forming the structures such as the conductive features 122 are intended to be illustrative and are not intended to be limiting upon the embodiments. Rather, any suitable methods and structures may be utilized to form the conductive features 122 and surrounding structures while still remaining within the scope of the embodiments. For example, another process that may be utilized can be found within U.S. Pat. No. 11,164,948, issued on Nov. 2, 2021, which reference is hereby incorporated herein by reference in its entirety. All such process and structures are fully intended to be included within the scope of the embodiments.

[0055] In FIGS. 18A and 18B, an etch stop layer 124 is selectively deposited over and in physical contact with the liner 121 and the conductive features 122, but not over dielectric materials such as the capping layers 100, the gate spacers 86, the CESL 87, and the first ILD 88. As discussed in greater detail below, a second ILD 126 (shown in FIGS. 19A-19B) will be formed over the etch stop layer 124 and a conductive feature 142 (shown in FIGS. 24A-24B) will be formed through the second ILD 126 to contact the conductive feature 122, wherein the etch stop layer 124 will act as an etch stop layer during the etch process. The etch stop layer 124 may be formed of a dielectric material having a lower etch rate than the subsequently formed second ILD 126 (based on the particular etchant chosen). As such, while the precise material used may be at least somewhat depen-

dent upon the material chosen for the second ILD 126, in some embodiments, the dielectric material of the etch stop layer 124 may be a nitride or an oxide of a metal material, such as aluminum oxide, aluminum nitride, tungsten nitride, molybdenum oxide, molybdenum nitride, boron nitride, or the like. However, any suitable material may be used.

[0056] The etch stop layer 124 is selectively deposited such that the etch stop layer 124 covers and is in physical contact with the conductive features 122 and/or the liner 121. In an embodiment in which the etch stop layer 124 is aluminum oxide and the conductive features are either tungsten or cobalt, the selective deposition may be performed as an atomic layer deposition process wherein the precursors are chosen to specifically react with the exposed portions of the conductive features 122 and/or the liner 121 but not react with the surrounding dielectric materials.

[0057] For example, in some embodiments, the etch stop layer 124 may be formed using an atomic layer deposition technique which utilizes a sequence of process steps whereby one monolayer of material is deposited. Such a sequence is referred to as a reaction cycle, wherein each process step is referred to as a pulse. Multiple reaction cycles are performed till the desired amount of material has been deposited. Each reaction cycle comprises a first reaction pulse using a first precursor and a second reaction pulse using a second precursor, and so on, depending on the number of reaction pulses required.

[0058] A purge pulse is performed after each reaction pulse to clear the processing chamber of any residual reactants and by-products prior to feeding further reactants into the processing chamber for the next reaction pulse. The purge between two successive reaction pulses may be performed using gases, such as He, Ar, H₂, N₂, N₂O, or NO, or any combination thereof, with a flow rate from about 100 sccm to about 10000 sccm. It is understood that the purge gas used in each purge pulse may be a different purge gas or the same purge gas. For example, if two purge pulses are performed in each reaction cycle then the first purge pulse (performed after the first reaction pulse) may use a first purge gas, and the second purge pulse (performed after the second reaction pulse) may use a second purge gas, the second purge gas may be a different purge gas or the same purge gas as the first purge gas.

[0059] In a particular embodiment, the selective deposition process may utilize a first precursor to supply the desired metal and a second precursor to supply the desired oxygen or nitrogen. For example, in an embodiment in which the desired metal is aluminum, the first precursor may be a precursor such as trimethylaluminum (TMA), tris (dimethylamido) aluminum (TDMAA), combinations of these, or the like. Further, in an embodiment in which the desired material is a metal oxide, the second precursor may be a precursor such as O₂, O₃, H₂O, alkyl alcohol compound having between 1 and 10 total carbon atoms, and/or aromatic alcohol compound having between 6 and 20 carbon atoms. However, any suitable precursor may be utilized.

[0060] In another particular embodiment in which a metal nitride is desired to be deposited as the etch stop layer 124, the first precursor may still be a precursor such as trimethylaluminum (TMA), tris (dimethylamido) aluminum (TDMAA), combinations of these, or the like. In this embodiment in which a nitride is desired, however, the second precursor would supply nitrogen atoms. As such, the second precursor may be a precursor such as diatomic nitrogen (N₂),

ammonia, combinations of these, or the like. Such a combination of precursors will deposit aluminum nitride instead of aluminum oxide.

[0061] In embodiments in which an atomic layer deposition process is utilized, the atomic layer deposition process is performed at temperatures and pressure suitable to deposit the desired material. For example, in some embodiments the atomic layer deposition process may be performed at a temperature from about 30° C. to about 600° C. Additionally, the pressure during the ALD deposition process may be from about 0.1 Torr to about 100 Torr. However, any suitable process parameters may be utilized.

[0062] In embodiments in which the selective deposition process is an atomic layer deposition process, the atomic layer deposition process cycles are continued until a desired thickness for the etch stop layer **124** is reached. In an embodiment the etch stop layer **124** can be formed to a thickness of between about 0.3 Å and about 100 Å, which may be achieved with a number of cycles of between about 1 and about 100. However, any suitable thickness and number of cycles may be utilized.

[0063] In FIGS. **19A** and **19B**, a second ILD **126** is formed over the first ILD **88**, the capping layers **100**, and the etch stop layers **124**. However, because the etch stop layers **124** are formed only over the conductive features **122** and/or the liners **121**, the second ILD **126** is deposited in physical contact with the capping layers **100**, the CESL **87**, and the first ILD **88**. In some embodiments, the second ILD **126** may be a material similar to that of the first ILD **88**, and may be formed in a similar manner. For example, the second ILD **126** may be formed of a dielectric material such as an oxide (e.g., silicon dioxide), PSG, BSG, BPSG, USG, or the like, and may be deposited by any suitable method, such as CVD, PECVD, or FCVD. The second ILD **126** may have various top and bottom surface profiles, which is discussed in greater detail later with respect to FIGS. **28A**, **28B**, and **28C**. An annealing process may be performed after the second ILD **126** is formed and, if desired, a planarization process such as a chemical mechanical polishing process may be performed.

[0064] In some embodiments, the material of etch stop layers **124** and/or the material of the second ILD **126** are chosen such that the etch rate of etch stop layers **124** is less than the etch rate of the second ILD **126** during an etch process of subsequently forming openings **134** and **135** (shown in FIGS. **22A-22B**) through the second ILD **126**. Because of the etch stop layers **124**, the second ILD **126** may be formed over the first ILD **88**, the capping layers **100**, and the conductive features **122** without depositing a blanket etch stop layer, thereby allowing for a thinner overall device.

[0065] FIGS. **20A** and **20B** illustrate a patterning of the second ILD **126** and the capping layers **100** to form openings **130** and **131**, in accordance with some embodiments. The openings **130** and **131** extend through the second ILD **126** and the capping layers **100** to expose top surfaces of the gate contact layers **101** over the gate electrodes **94**. The second ILD **126** and the capping layers **100** may be patterned using acceptable photolithography and etching techniques. For example, a first photoresist **128** may be formed over the second ILD **126** and patterned using suitable photolithography techniques. The first photoresist **128** may be a single layer or multilayer photoresist structure, and may be deposited using suitable techniques such as spin-on or deposition techniques. One or more suitable etch processes may then be performed using the patterned first photoresist **128** as an etch

mask and the capping layers **100** as an etch stop layer while etching through the second ILD **126**, thereby forming the openings **130** and **131**. The one or more etch processes may include wet and/or dry etch processes. FIGS. **20A** and **20B** show the openings **130** and **131** as having sloped sidewalls, but the openings **130** or **131** may have substantially vertical sidewalls, curved sidewalls, or another sidewall profile in other embodiments. The first photoresist **128** may be removed using a suitable process such as an ashing or etch process.

[0066] As discussed previously, by forming etch stop layers **124** over the conductive features **122**, the second ILD **126** may be formed over the first ILD **88**, the capping layers **100**, and the conductive features **122** without depositing a blanket etch stop layer, thereby reducing a thickness of the layers above the gate electrodes **94** that need to be etched when forming the openings **130** and **131**. As such, the etching process through the combined etch stop layer **124** and the capping layers **100** can be well controlled (because the thickness of the materials over the capping layers **100** is reduced), thereby improving the overall yield.

[0067] In FIGS. **21A** and **21B**, a second photoresist **132** is formed over the second ILD **126** and within the openings **130** and **131**, in accordance with some embodiments. The second photoresist **132** may be a single layer or multilayer photoresist structure, and may be deposited using suitable techniques such as spin-on or deposition techniques. As shown in FIG. **21B**, the second photoresist **132** may overfill the openings **130** and **131** and extend over the second ILD **126**.

[0068] FIGS. **22A** and **22B** illustrate the patterning of the second photoresist **132**, the second ILD **126**, and the etch stop layers **124** to form openings **134** and **135**, in accordance with some embodiments. The openings **134** and **135** extend through the second ILD **126** and the etch stop layers **124** to expose the top surfaces of the conductive features **122**. The second photoresist **132**, the second ILD **126** and the etch stop layers **124** may be patterned using acceptable photolithography and etching techniques. One or more suitable etch processes may then be performed using the patterned second photoresist **132** as an etch mask, forming the openings **134** and **135**. The one or more etch processes may include wet and/or dry etch processes.

[0069] In some embodiments, the etch process(es) may remove portions of the second ILD **126** and then stop or slow down at the etch stop layers **124**, which reduces the chance of over-etching the conductive features **122**, and therefore reduces the chance of forming leakage paths or other defects. A separate etch process may be performed to remove the portions of the etch stop layers **124** and expose the top surfaces of the conductive features **122**. The separate etch process may utilize different etchants from the etch process(es) used to etch the second ILD **126**. In some embodiments, the openings **134** or **135** may expose sidewalls of the etch stop layers **124**, sidewalls of the first ILD **88**, and/or sidewalls of the CESL **87**. FIGS. **22A** and **22B** show the openings **134** and **135** as having sloped sidewalls, but the openings **134** or **135** may have substantially vertical sidewalls, curved sidewalls, or another sidewall profile in other embodiments.

[0070] In some embodiments, the etch process(es) may form an opening for a butted contact, e.g., a joint or combined contact to the conductive feature **122** (coupled to the epitaxial source/drain region **82**) and an adjacent gate

electrode 94. For example, the opening 135 illustrated in FIG. 22B illustrates an example in which the opening 135 overlaps the second photoresist 132 above the gate electrode 94 on the left side of the figure. In subsequent processing, the second photoresist 132 is removed, thereby forming an opening exposing both the gate electrode 94 and the conductive feature 122. The etch rate of the materials of the various layers (e.g., the gate seal spacer 80, the gate spacer 86, the CESL 87, the first ILD 88, the capping layer 100, and/or the second photoresist 132) may be different from the etch rate of the material of the etch stop layer 124. The difference in the etch rates may cause an upper surface of the gate seal spacer 80, the gate spacer 86, the CESL 87, the first ILD 88, the capping layer 100, and/or the second photoresist 132 to be the same, higher or lower than the upper surface of the conductive feature 122. FIG. 22B illustrates an example of the etch rate of the gate seal spacer 80, the gate spacer 86, the CESL 87, the first ILD 88, the capping layer 100, and/or the second photoresist 132 being less than the etch rate of the etch stop layer 124.

[0071] In FIGS. 23A and 23B, the second photoresist 132 is removed, forming openings 130, 134, and 136, in accordance with some embodiments. The second photoresist 132 may be removed using a suitable technique, such as by ashing, etching, or the like. As shown in FIGS. 23A and 23B, removing the second photoresist 132 reveals the previously formed opening 130 that exposes the top surface of the gate contact layer 101 and the gate electrode 94. Due to the overlap between the previously formed openings 131 and 135, removing the second photoresist 132 forms a combined opening 136 that exposes the top surface of the gate electrode 94 or the gate contact layer 101 previously exposed by the opening 131 and the top surface of conductive feature 122 previously exposed by the opening 135. The opening 134 remains exposing the top surface of the conductive feature 122. In some embodiments, a wet cleaning process is performed before and/or after removing the second photoresist 132.

[0072] In FIGS. 24A and 24B, a conductive feature 140, a conductive feature 142, and a combined conductive feature 144 (e.g., a butted contact) are formed respectively in the opening 130, the opening 134, and the combined opening 136. In some embodiments, forming the conductive features 140, 142 and 144 may comprise forming a liner (not shown), such as a barrier layer, an adhesion layer, or the like, and a conductive material in the openings 130, 134, and 136. The liner may include titanium, titanium nitride, tantalum, tantalum nitride, or the like. The conductive material may be copper, a copper alloy, silver, gold, tungsten, cobalt, aluminum, nickel, or the like. A planarization process, such as a CMP, may be performed to remove excess material from a surface of the second ILD 126. The remaining liner and conductive material form the conductive features 140, 142 and 144. The conductive features 140, 142 and 144 may be formed in different processes or may be formed in the same process. Although shown as being formed in the same cross-sections, it should be appreciated that the conductive features 140, 142 and/or 144 may be formed in different cross-sections, which may avoid/reduce a risk of shorting.

[0073] The conductive feature 140 makes electrical connection to the gate electrode 94. Accordingly, the conductive feature 140 may be referred to as a gate contact or gate contact plug in some cases. The conductive feature 142 makes electrical connection to the conductive feature 122

that is electrically connected to the epitaxial source/drain region 82. Accordingly, a combination of the conductive feature 142 and the underlying conductive feature 122 may be also referred to as a source/drain contact or a source/drain contact plug in some cases. The combined conductive feature 144 is electrically connected to both the gate electrode 94 and the epitaxial source/drain region 82 (through one of the conductive features 122). In this manner, a FinFET device comprising gate contact plugs and source/drain contact plugs may be formed. As discussed previously, by forming etch stop layers 124 over the conductive features 122, the second ILD 126 may be formed over the first ILD 88, the capping layers 100, and the conductive features 122 without depositing a blanket etch stop layer, which may provide better profiles of the conductive feature 140 and the combined conductive feature 144, and lead to a better electrical connection.

[0074] In FIGS. 25A and 25B, an interconnect structure comprising one or more layers of conductive features are formed over and electrically connected to the conductive features 140, 142, and 144. In some embodiments, the interconnect structure comprises a plurality of dielectric layers such as inter-metal dielectric layers (IMDs) and conductive features within the IMDs that provide various electrical interconnections. FIGS. 25A and 25B illustrate an embodiment comprising one IMD 152 with conductive features 150 and one IMD 155 with conductive features 154, but more or fewer IMDs or conductive features may be formed in other embodiments. The conductive features 150 and 154 may comprise electrical routing, conductive vias, conductive lines, or the like, and may be formed using a single damascene method, a dual damascene method, a combination thereof, or the like.

[0075] In some embodiments, an etch stop layer 151 may first be deposited over the second ILD 126 and conductive features 140, 142, and 144. The etch stop layer 151 may comprise a material such as silicon nitride, silicon oxynitride, aluminum oxide, aluminum nitride, the like, or combinations thereof. Other materials are possible. The IMD 152 may then be formed over the etch stop layer 151. The IMD 152 may be a material similar to that described for the first ILD 88 or the second ILD 126, and may be formed in a similar manner. In some embodiments, the IMD 152 may be formed of a low-k dielectric material having a k-value lower than about 3.5. Other materials or techniques are possible. Openings may then be patterned in the IMD 152 and the etch stop layer 151 to expose surfaces of the conductive features 140, 142, and/or 144. An optional liner (not shown) may first be formed in the openings, which may be similar to the liner described previously for the conductive features 140, 142, and 144. A conductive material may be deposited within the openings to form the conductive features 150. The conductive material may be similar to those described for the conductive features 140, 142, and 144, and may be formed in a similar manner. Other conductive materials or techniques are possible. A planarization process may be performed to remove excess conductive material from the IMD 152. FIGS. 25A and 25B show the conductive features 150 as having sloped sidewalls, but the conductive features 150 may have substantially vertical sidewalls, curved sidewalls, or another sidewall profile in other embodiments.

[0076] The conductive features 154 may be formed in a similar manner as the conductive features 150. For example,

an etch stop layer 153 may be formed over the IMD 152 and conductive features 150, and the IMD 155 may be formed over the IMD 152. The etch stop layer 153 and the IMD 152 may be patterned to form openings. Some of the openings may expose the conductive features 150. A liner and a conductive material may then be deposited in the openings, and a CMP process may be performed to remove excess materials. FIGS. 25A and 25B show the conductive features 154 as having sloped sidewalls, but the conductive features 154 may have substantially vertical sidewalls, curved sidewalls, or another sidewall profile in other embodiments.

[0077] Certain features of the FinFET embodiments discussed herein could also be applied to nanostructure devices such as NFETs/NSFETs. As an example, FIGS. 26A and 26B are cross-sectional views of a NSFET device, in accordance with some embodiments. The NSFET device is similar to the FinFET device shown in FIGS. 25A and 25B. As such, similar features in FIGS. 25A-25B and FIGS. 26A-26B are labeled by similar numerical references. The channel regions of the NSFET device comprise nanostructures 160 separated from the fins 52 and encircled by the respective gate stacks (e.g., the gate dielectric layers 92 and the gate electrodes 94), as shown in FIG. 26A. The nanostructures 160 may comprise nanosheets, nanowires, or the like. The nanostructures 160 and the substrate 50 may comprise a similar semiconductor material or different semiconductor materials. In some embodiments, portions of the gate stacks are interposed between the adjacent nanostructures 160 and spacers 162 are interposed between the portions of the gate stacks and the epitaxial source/drain regions 82, as shown in 26B. The spacers 162 may act as isolation features between the gate stacks and the epitaxial source/drain regions 82. In some embodiments, the spacers 162 comprise a material such as silicon nitride or silicon oxynitride, although any suitable material, such as a low-k dielectric material, may be utilized. The conductive features 140, 142, and 144 contacting the top surfaces of the gate electrodes and the top surfaces of the conductive features 122 may be formed in a manner similar to the FinFET embodiments discussed previously. In some embodiments, an etch stop layer 124 may be formed on the top surfaces of the conductive features 122, which may be formed in a similar manner and serves a similar purpose as the FinFET embodiments discussed previously.

[0078] FIGS. 27A, 27B, and 27C illustrate various top surface profiles of the conductive features 122 and, given the selective deposition process, top surface profiles of the etch stop layers 124 formed thereon in accordance with some embodiments. The structures shown in FIGS. 27A, 27B, and 27C are portions of the structure shown in FIGS. 18B, wherein top surfaces of the conductive features 122 and the etch stop layer 124 have different top surface profiles. For example, FIG. 27A illustrates an etch stop layer 124 with a flat top surface (the same as seen in FIG. 18B), wherein the top surfaces of the conductive features 122, the first ILD 88, the CESL 87, and the capping layers 100 may be level. In some cases, forming the etch stop layer 124 with a flat top surface or forming the etch stop layer 124 with uniform thickness leads to a better controlled etch process when the etch stop layer 124 is etched through, thereby resulting in a better profile of the conductive feature 142, as shown in FIG. 24B. FIG. 27B illustrates an etch stop layer 124 with a convex or protruding top surface. FIG. 27C illustrates an etch stop layer 124 with a concave or dished top surface such

that the top surface of the etch stop layer 124 potentially extends below the top surfaces of the first ILD 88, the CESL 87, and the capping layers 100.

[0079] In some embodiments, the top surface profile of the etch stop layer 124 may be controlled by controlling the polishing rate of the conductive features 122 (e.g., R1) and the polishing rate of the surrounding layers, such as the first ILD 88 (e.g., R2), the CESL 87, and/or the capping layers 100. In some embodiments in which R1 is the same as R2, the planarization process and subsequent deposition process may result in the etch stop layer 124 with a flat top surface, such as illustrated in FIG. 27A. In other embodiments R1 is smaller than R2, and the planarization process and subsequent deposition process may result in the etch stop layer 124 with a convex or protruding top surface, such as illustrated in FIG. 27B. In still other embodiments, R1 is greater than R2, and the planarization process and subsequent deposition process may result in the etch stop layer 124 with a concave or dished top surface, such as illustrated in FIG. 27C.

[0080] FIGS. 28A, 28B, and 28C show the second ILD 126 formed on the etch stop layers 124 of FIGS. 27A, 27B, and 27C, in accordance with some embodiments. The structures shown in FIGS. 28A, 28B, and 28C are portions of the structure shown in FIG. 19B, wherein an upper surface of the second ILD 126 corresponds to an upper surface of the underlying etch stop layer 124. FIG. 28A shows a second ILD 126 formed on the top surface of the etch stop layer 124 of FIG. 27A, in accordance with some embodiments. The second ILD 126 may have a generally flat top surface with a bulge where the etch stop layer 124 extends upwards from the conductive feature 122. FIG. 28B shows a second ILD 126 formed on the convex or protruding top surface of the etch stop layer 124 of FIG. 27B, in accordance with some embodiments. The portion of the top surface of the second ILD 126 over the etch stop layer 124 may be convex or protruding and the portion of the bottom surface of the second ILD 126 over the etch stop layer 124 may be concave or dishing. FIG. 28C shows a second ILD 126 formed on the concave or dishing top surface of the etch stop layer 124 of FIG. 27C, in accordance with some embodiments. The portion of the top surface of the second ILD 126 over the etch stop layer 124 may be concave or dishing and the portion of the bottom surface of the second ILD 126 over the etch stop layer 124 may be convex or protruding.

[0081] The embodiments described herein have some advantages. For example, an etch stop layer may be formed selectively on the conductive feature connected to the source/drain region instead of depositing a blanket etch stop layer over the source/drain region and the gate stack. By forming the etch stop layer selectively, the etch stop layer is not formed on surrounding dielectric materials, and subsequent etching processes can have a smaller pattern loading effect, thereby enlarging the over/under etch free process window as well as reducing resistance. As such, the etching process can be well controlled, leading to an overall improvement in yield.

[0082] In accordance with an embodiment, a method of manufacturing a semiconductor device includes: depositing a source/drain contact in electrical connection with a source/drain region over a semiconductor fin; selectively depositing an etch stop layer on the source/drain contact; depositing a dielectric layer over the etch stop layer, wherein after the depositing the dielectric layer the dielectric layer is in

physical contact with a gate capping layer overlying a gate electrode; forming a first conductive contact to the gate electrode through the gate capping layer; and forming a second conductive contact to the source/drain contact through the etch stop layer. In an embodiment, the selectively depositing the etch stop layer is performed at least in part with an atomic layer deposition process. In an embodiment, the selectively depositing the etch stop layer deposits aluminum oxide. In an embodiment, the selectively depositing the etch stop layer deposits the etch stop layer on a liner adjacent to the source/drain contact. In an embodiment, the method further includes forming a butted contact at least partially through the etch stop layer. In an embodiment, a barrier layer adjacent to the source/drain contact extends a first distance away from the semiconductor fin, the gate electrode extends a second distance less than the first distance away from the semiconductor fin, and a spacer adjacent to the gate electrode extends a third distance away from the semiconductor fin, the third distance being between the first distance and the second distance. In an embodiment, the source/drain contact comprises tungsten.

[0083] In another embodiment, a method of manufacturing a semiconductor device includes: depositing an etch stop layer in physical contact with a source/drain contact, wherein the depositing the etch stop layer does not deposit the etch stop layer on one or more exposed dielectric materials; depositing a dielectric layer over the etch stop layer; forming a first opening through the dielectric layer and the etch stop layer to expose the source/drain contact; forming a second opening through the dielectric layer but not the etch stop layer to expose a first gate electrode; and filling the first opening and the second opening with a conductive material. In an embodiment, the forming the first opening further exposes a second gate electrode. In an embodiment, the depositing the etch stop layer uses an aluminum precursor. In an embodiment, the aluminum precursor comprises trimethylaluminum. In an embodiment, the aluminum precursor comprises tris (dimethylamido) aluminum. In an embodiment, the depositing the etch stop layer uses an oxygen precursor. In an embodiment, the depositing the etch stop layer uses a nitrogen precursor.

[0084] In accordance with yet another embodiment, a semiconductor device includes: a gate electrode overlying a semiconductor fin; a source/drain contact separated from the gate electrode by a dielectric material; an etch stop layer over the source/drain contact but not extending over the dielectric material; a dielectric layer overlying the etch stop layer and in physical contact with the dielectric material; a first conductive contact extending through the dielectric layer to make electrical contact with the gate electrode; and a second conductive contact extending through the dielectric layer and the etch stop layer to make electrical contact with the source/drain contact. In an embodiment, the etch stop layer comprises aluminum oxide. In an embodiment, the gate electrode extends a first distance away from the semiconductor fin and a barrier layer adjacent to the source/drain contact extends a second distance further than the first distance. In an embodiment, the semiconductor device further includes a spacer adjacent to the gate electrode, the spacer extending a third distance between the first distance and the second distance. In an embodiment, the etch stop layer comprises a metal nitride. In an embodiment, the etch stop layer comprises a metal oxide.

[0085] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. A semiconductor device comprising:
 - a source/drain contact in electrical connection with a source/drain region over a semiconductor fin, the source/drain contact being separated from a gate structure by a spacer;
 - an etch stop layer on the source/drain contact but not on the spacer;
 - a dielectric layer over the etch stop layer, wherein the dielectric layer is in physical contact with a gate capping layer overlying a gate electrode;
 - a first conductive contact to the gate electrode through the gate capping layer; and
 - a second conductive contact to the source/drain contact through the etch stop layer.
2. The semiconductor device of claim 1, wherein the etch stop layer is in physical contact with a liner.
3. The semiconductor device of claim 1, wherein the etch stop layer comprises aluminum oxide.
4. The semiconductor device of claim 1, wherein the etch stop layer comprises aluminum nitride.
5. The semiconductor device of claim 1, wherein the etch stop layer comprises tungsten nitride.
6. The semiconductor device of claim 1, wherein the etch stop layer comprises molybdenum oxide.
7. The semiconductor device of claim 1, wherein the etch stop layer comprises boron nitride.
8. A semiconductor device comprising:
 - an etch stop layer in physical contact with a source/drain contact, wherein the etch stop layer has a bottom surface facing a semiconductor substrate, the bottom surface not being in physical contact with one or more exposed dielectric materials;
 - a dielectric layer over the etch stop layer;
 - a first conductive material extending through the dielectric layer and the etch stop layer to the source/drain contact; and
 - a second conductive material extending through the dielectric layer but not the etch stop layer to a first gate electrode.
9. The semiconductor device of claim 8, wherein the first conductive material is in physical contact with a gate contact.
10. The semiconductor device of claim 8, wherein the etch stop layer has a thickness of between about 0.3 Å and about 100 Å.
11. The semiconductor device of claim 8, wherein there is no blanket etch stop layer between the dielectric layer and the source/drain contact.
12. The semiconductor device of claim 8, wherein the etch stop layer has a concave surface.

13. The semiconductor device of claim **8**, wherein the etch stop layer has a convex surface.

14. The semiconductor device of claim **8**, wherein the etch stop layer comprises molybdenum nitride.

15. A semiconductor device comprising:

a first gate structure over a semiconductor substrate;

a spacer adjacent to the first gate structure;

a source/drain located on an opposite side of the spacer from the first gate structure;

a source/drain contact in physical contact with the source/drain;

an etch stop layer in physical contact with the source/drain contact, wherein a first surface of the spacer is free from the etch stop layer, the first surface facing away from the semiconductor substrate;

a dielectric layer over the etch stop layer;

a first conductive material extending through the dielectric layer and the etch stop layer to the source/drain contact; and

a second conductive material extending through the dielectric layer but not the etch stop layer to a first gate electrode.

16. The semiconductor device of claim **15**, wherein there is no blanket etch stop layer between the dielectric layer and the source/drain contact.

17. The semiconductor device of claim **15**, wherein the first conductive material is a butted contact.

18. The semiconductor device of claim **15**, wherein the etch stop layer has a curved top surface.

19. The semiconductor device of claim **18**, wherein the curved top surface is concave.

20. The semiconductor device of claim **18**, wherein the curved top surface is convex.

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